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Kwak

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(54) **OPERATION METHOD OF UE RECEIVING V2X-RELATED MESSAGE IN WIRELESS COMMUNICATION SYSTEM**

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H04L 43/18 (2022.01)

H04W 92/18 (2009.01)

(52) **U.S. Cl.**

CPC **H04W 28/06** (2013.01); **H04L 43/18** (2013.01); **H04W 92/18** (2013.01)

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H04W 28/02; H04W 92/10; H04L 43/18;
H04L 43/028

See application file for complete search history.

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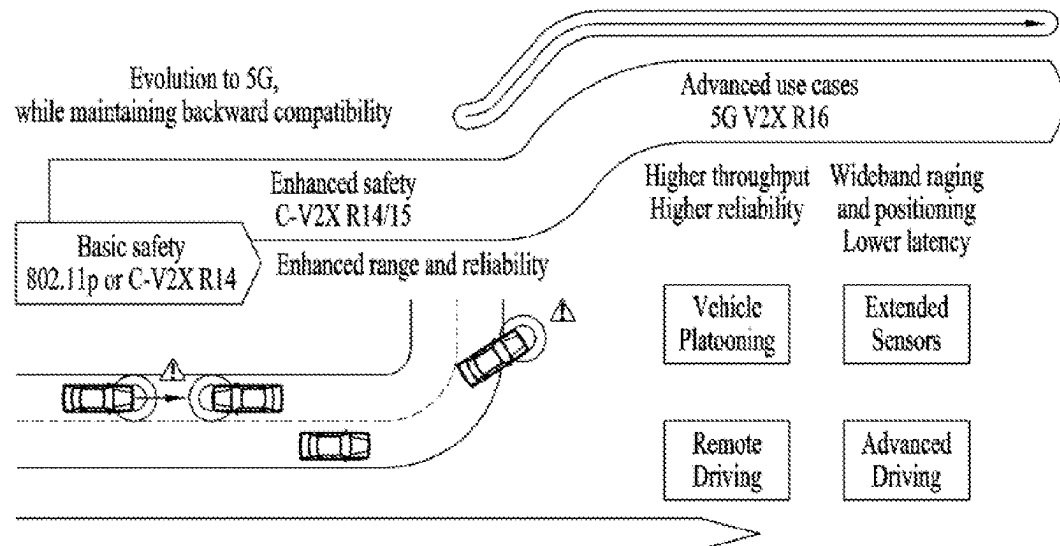
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(57) **ABSTRACT**

One embodiment provides an operation method of a first UE in a wireless communication system, the method comprising steps in which a first UE receives a V2X-related packet from either a second UE or a base station; and first UE acquires a V2X-related message on the basis of the parsing of the V2X-related packet, wherein the V2X-related packet includes header information for indicating whether the V2X related packet is a LTE-V2X PC5-related packet or a LTE-V2X Uu-interface-related packet.

10 Claims, 21 Drawing Sheets



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FIG. 1

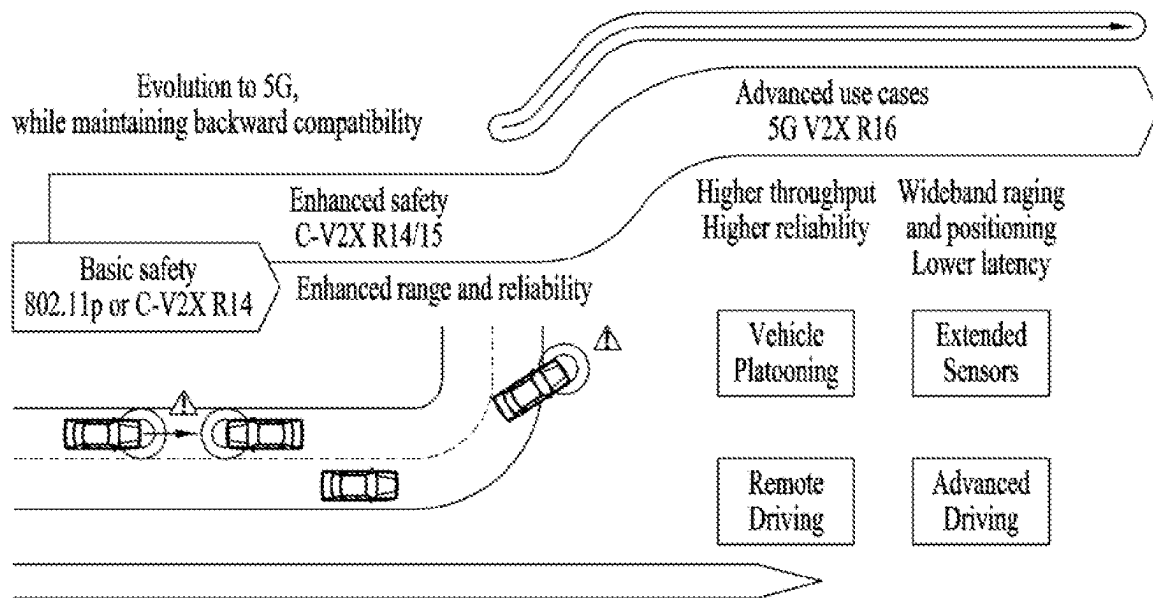


FIG. 2

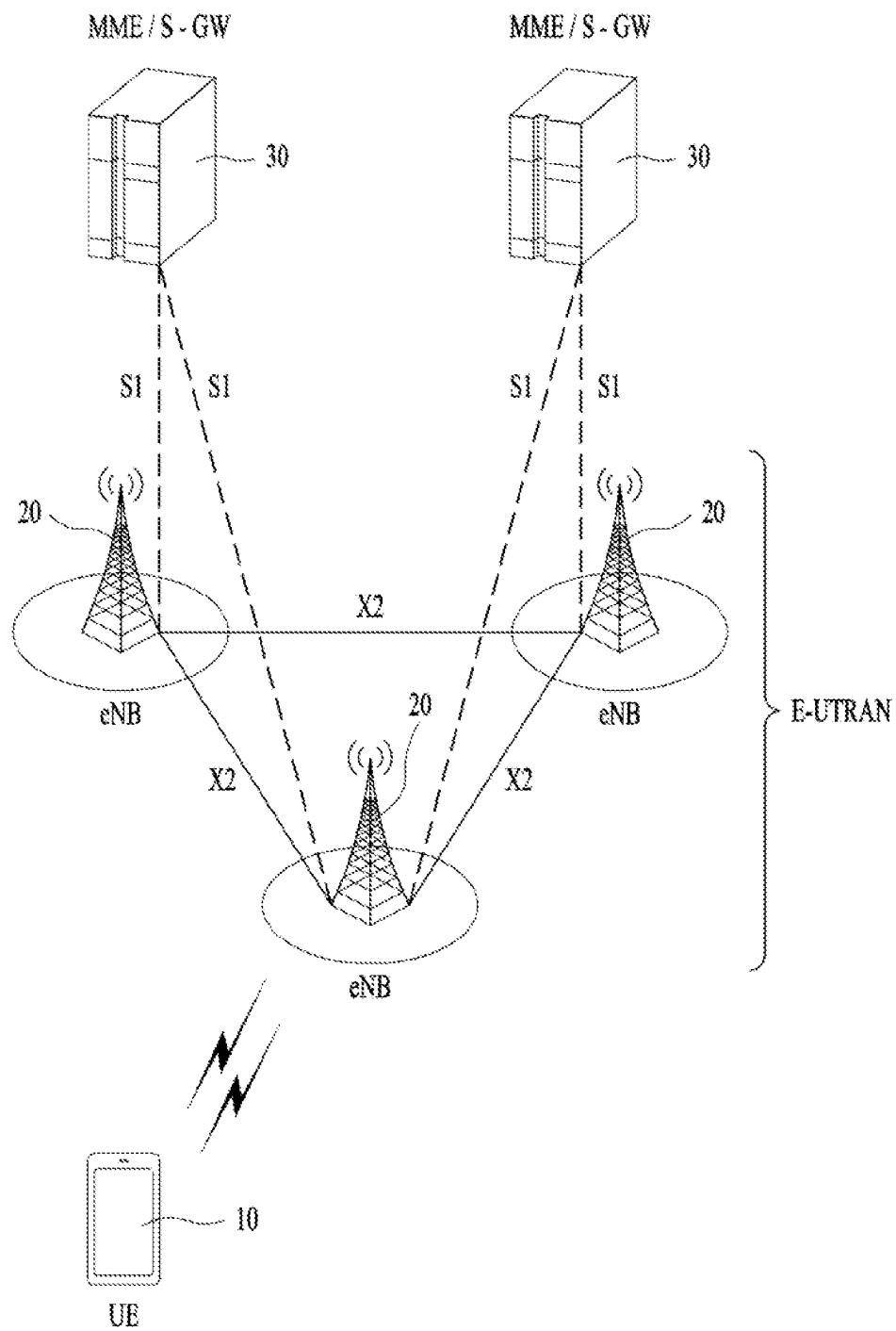
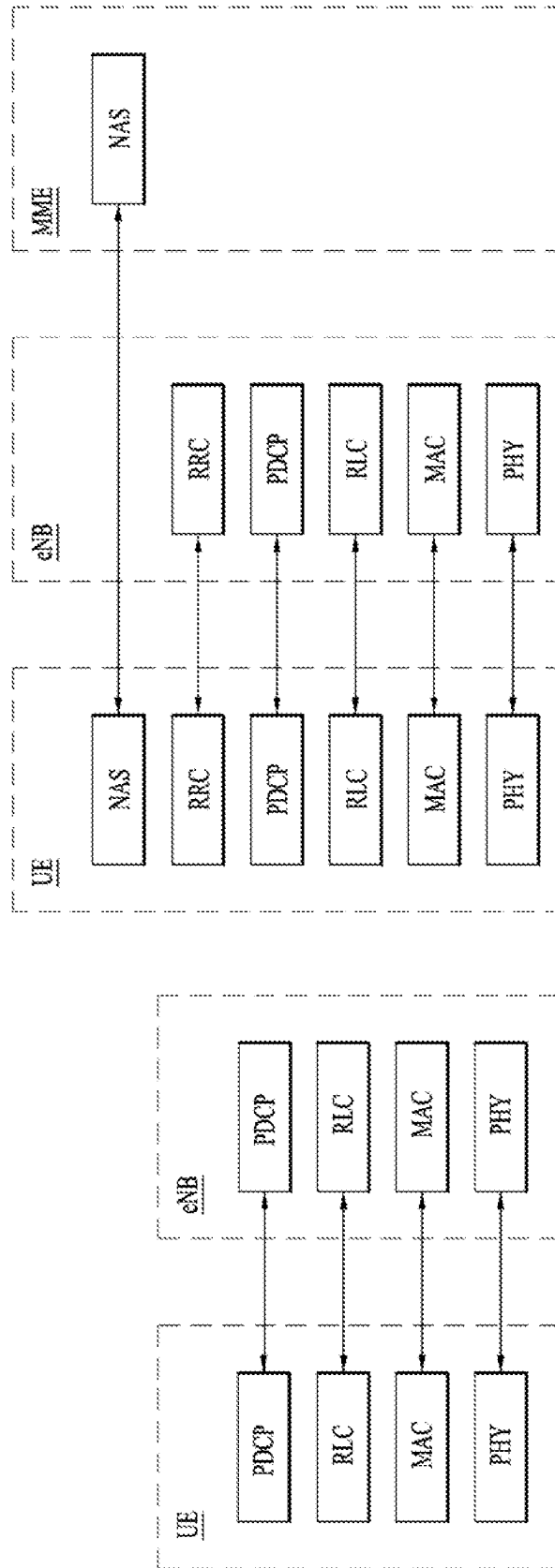


FIG. 3



(a)

(b)

FIG. 4

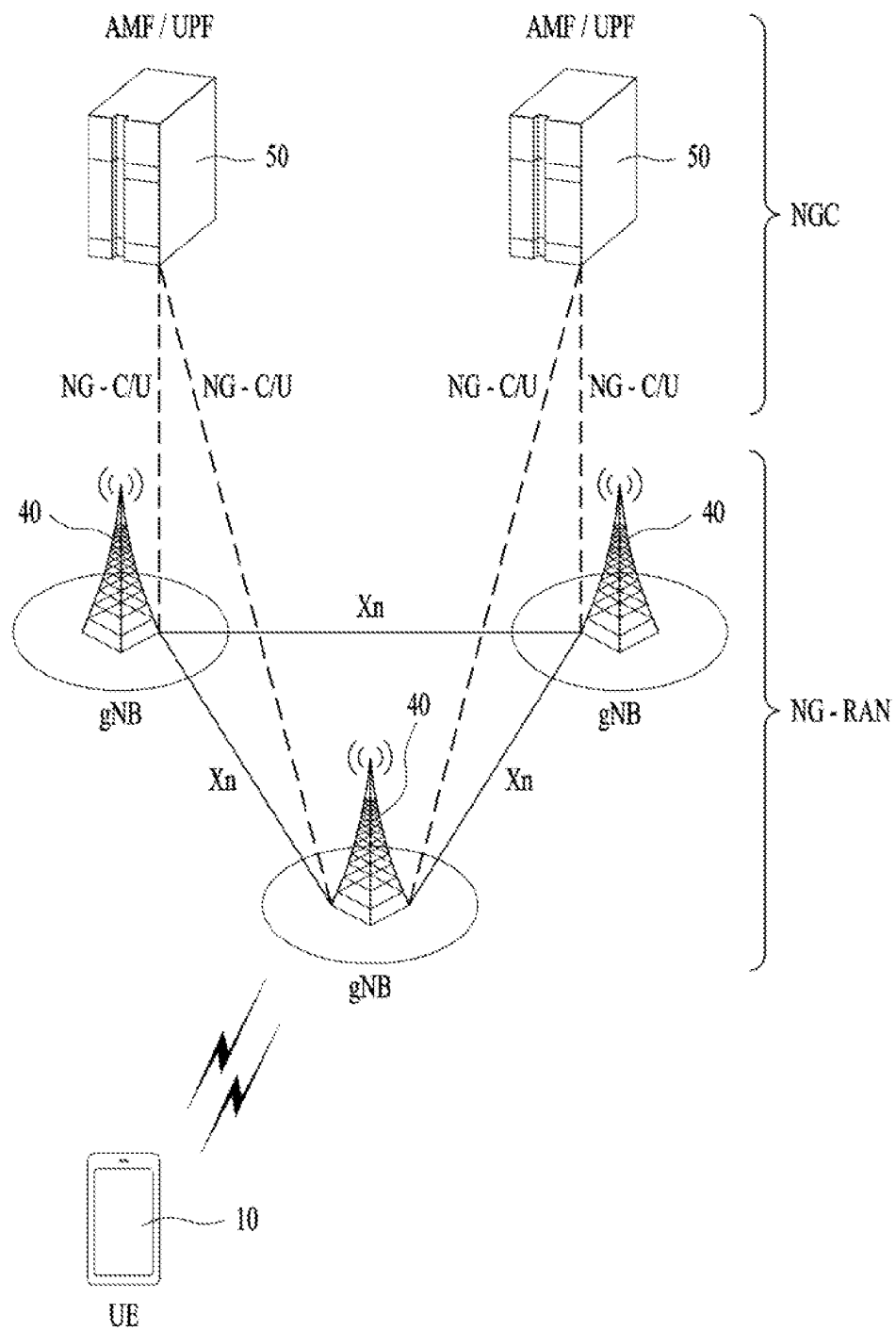


FIG. 5

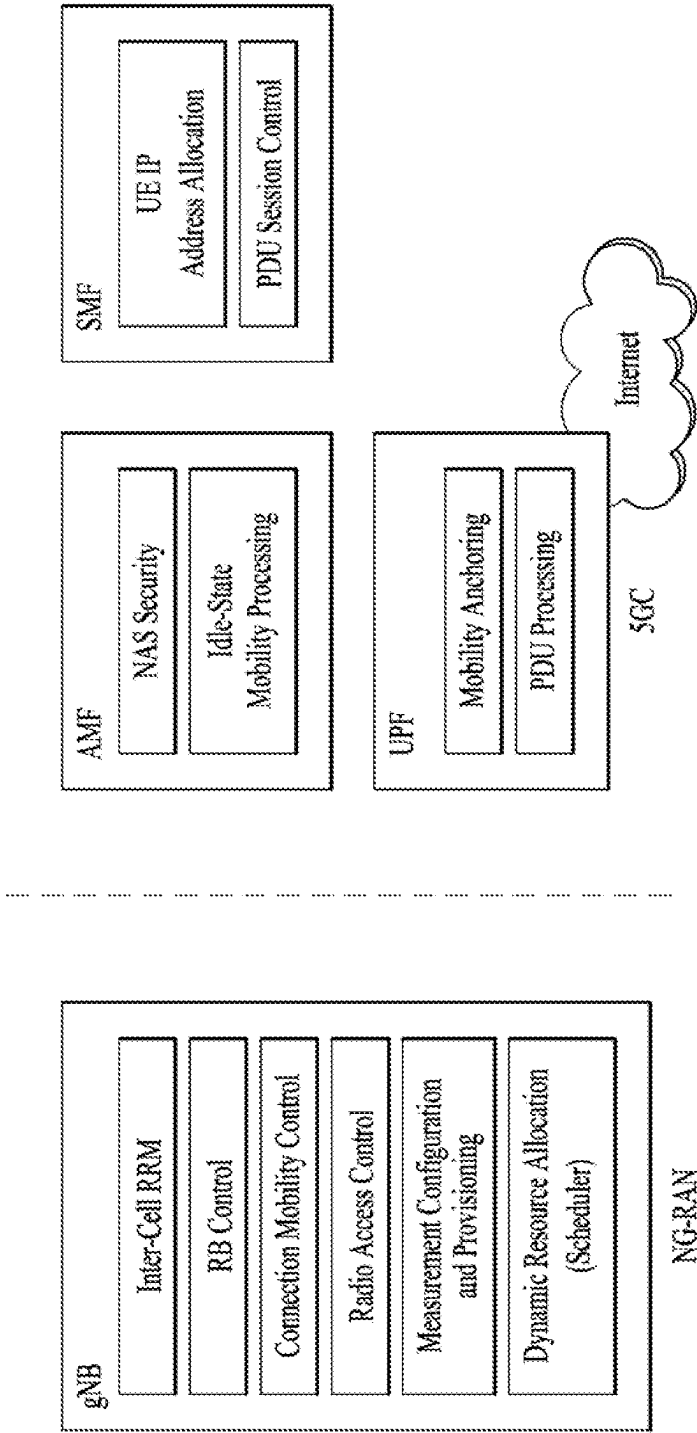


FIG. 6

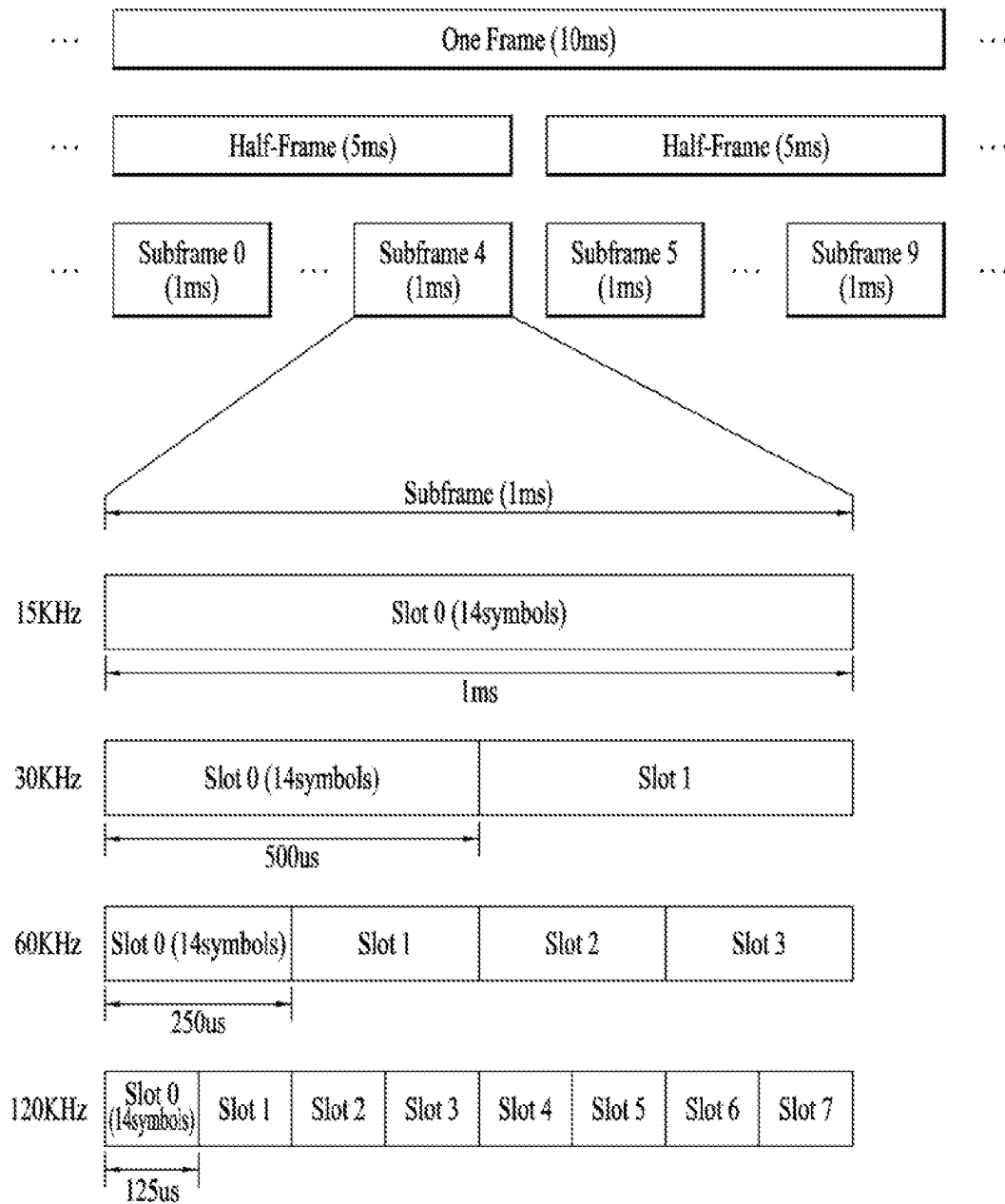


FIG. 7

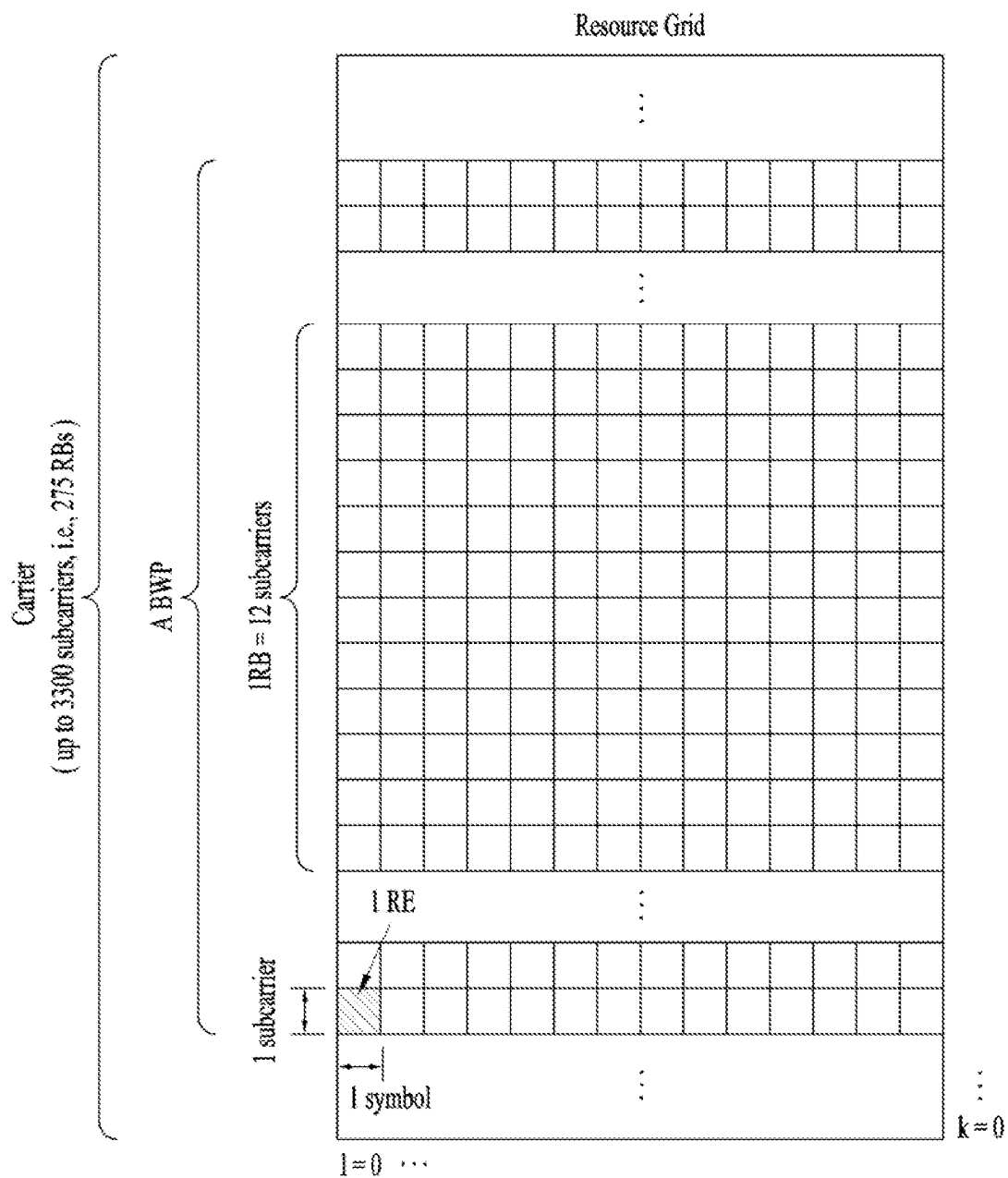


FIG. 8

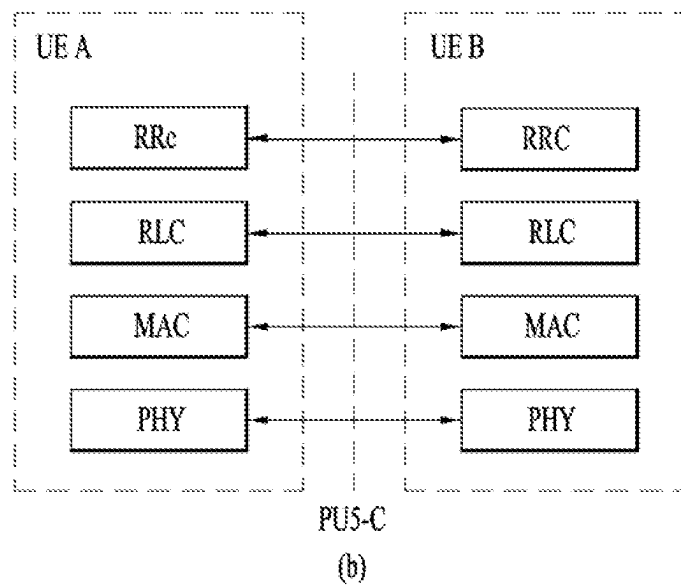
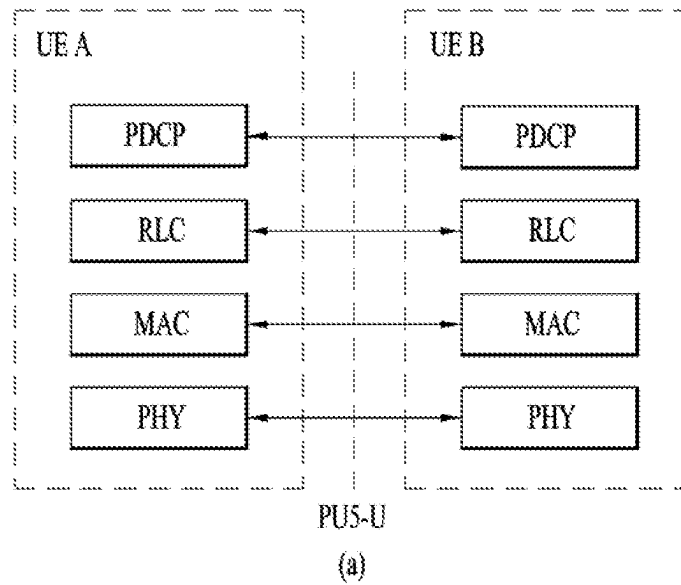


FIG. 9

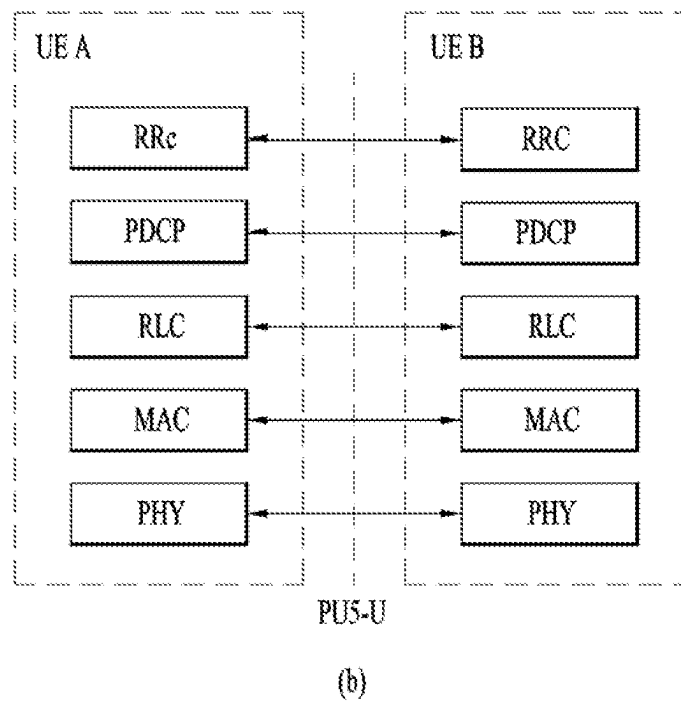
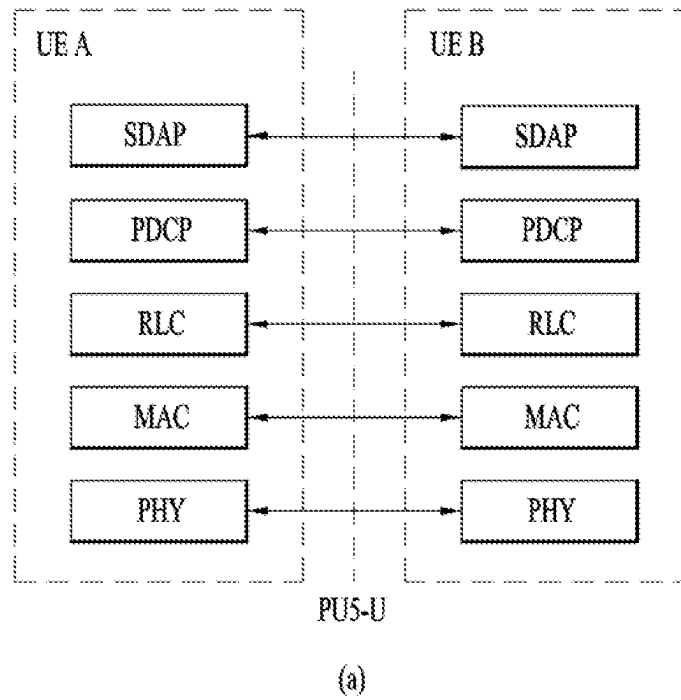


FIG. 10

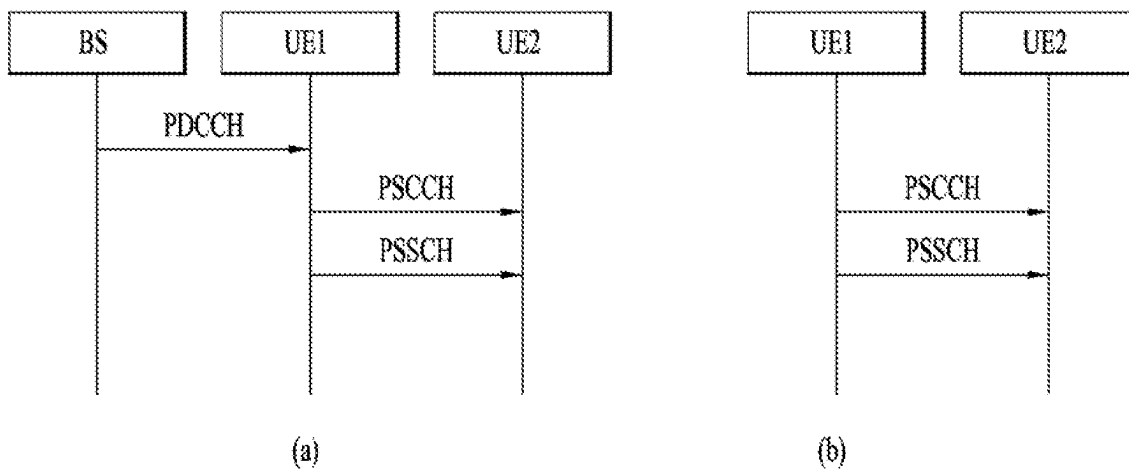


FIG. 11

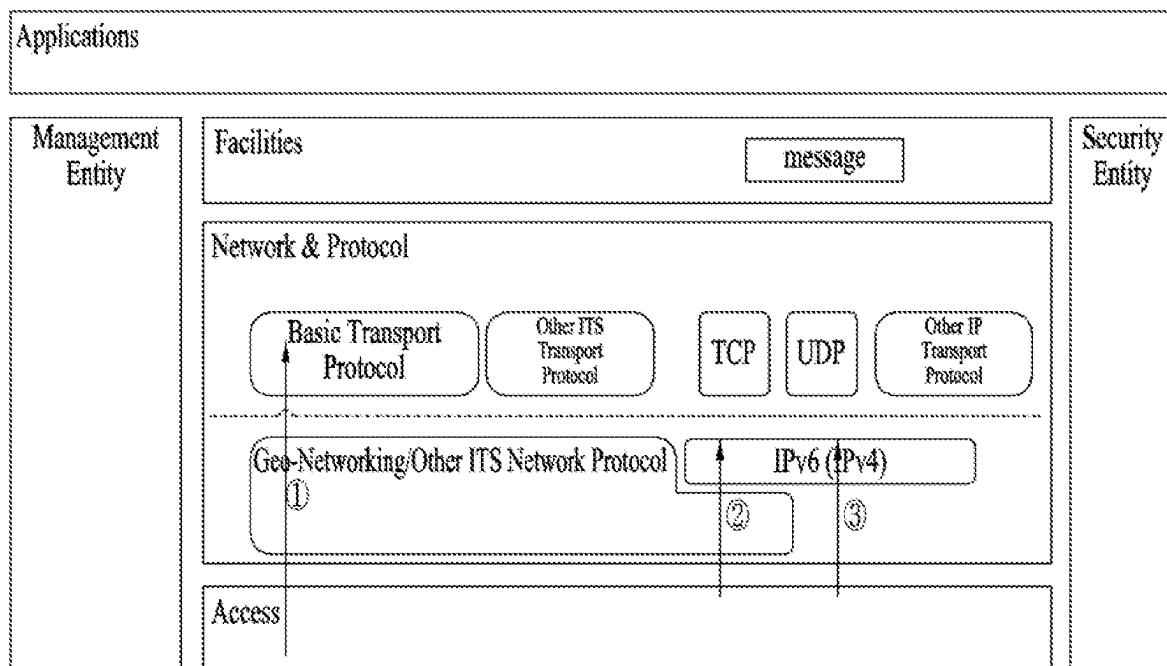
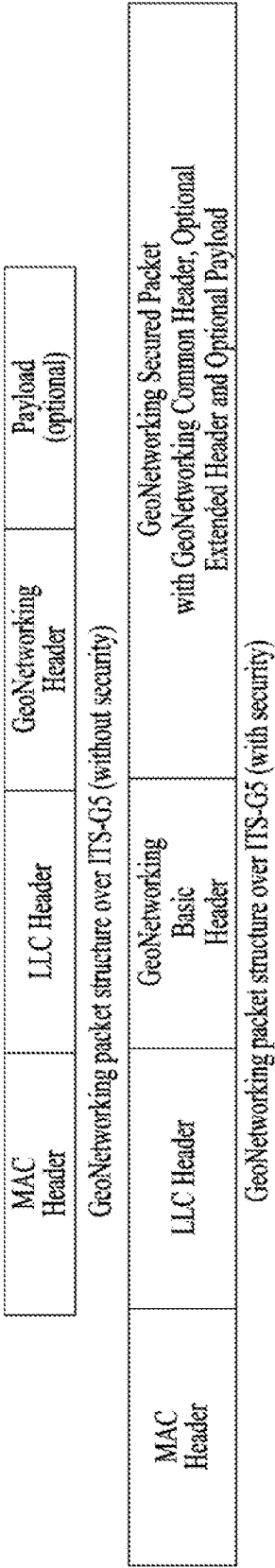


FIG. 12

1) ITS-G5 access layer technology



2) Overall packet structure for LTE-V2X access layer technology

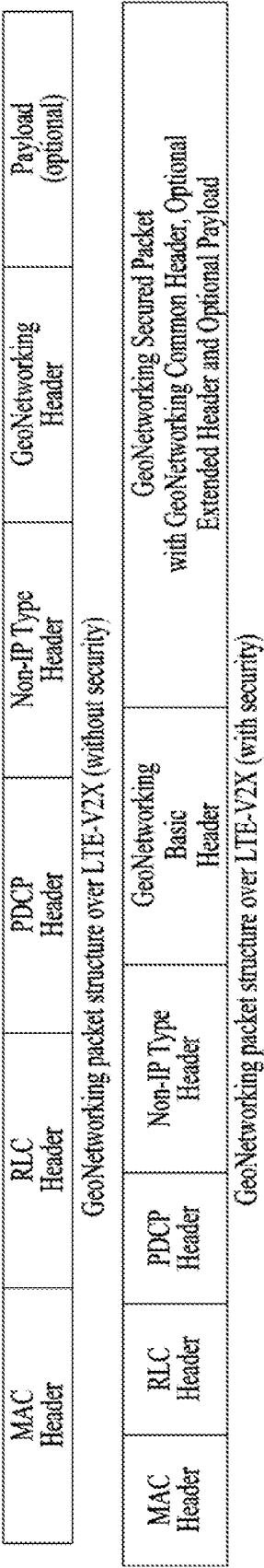


FIG. 13

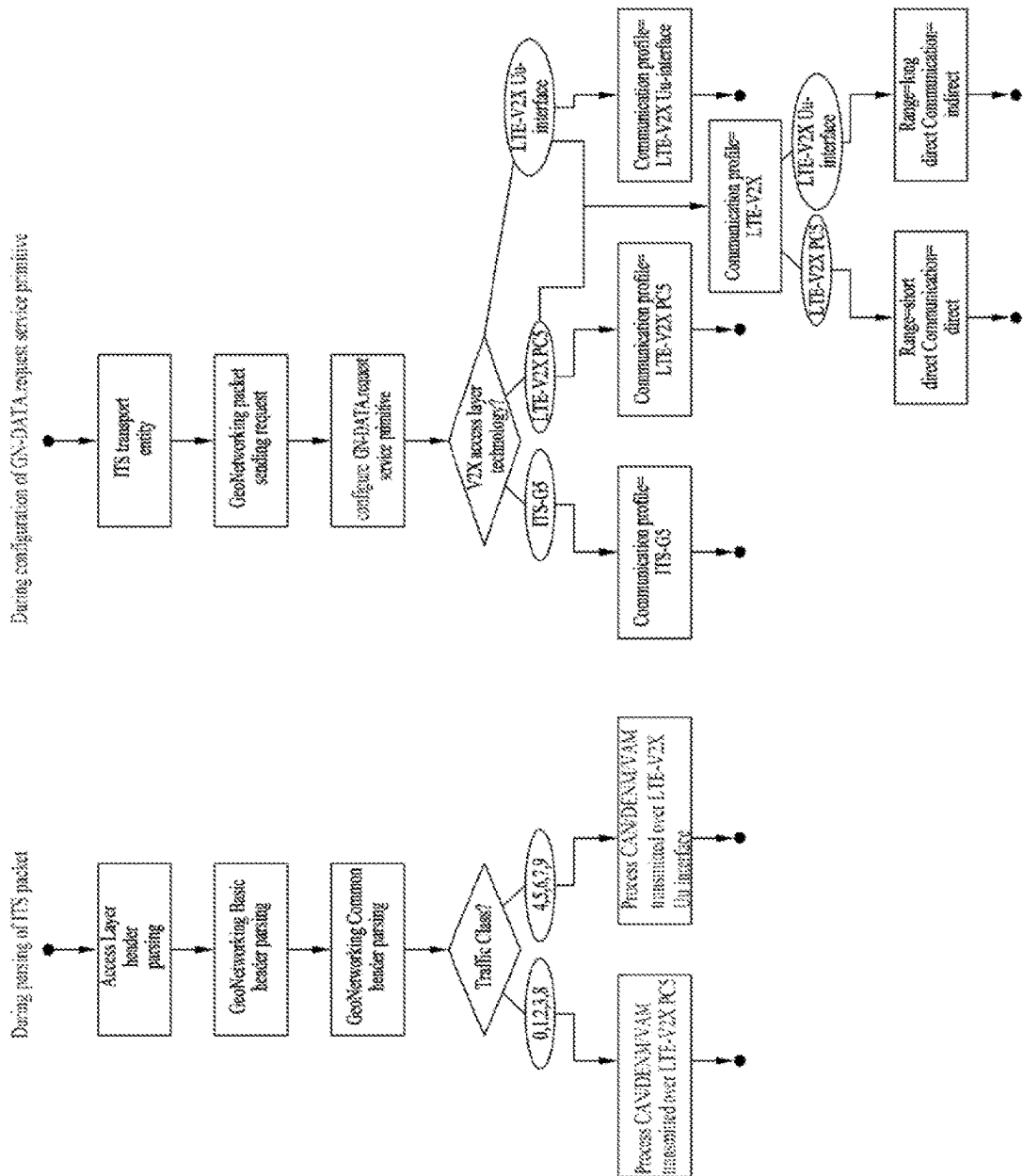


FIG. 14

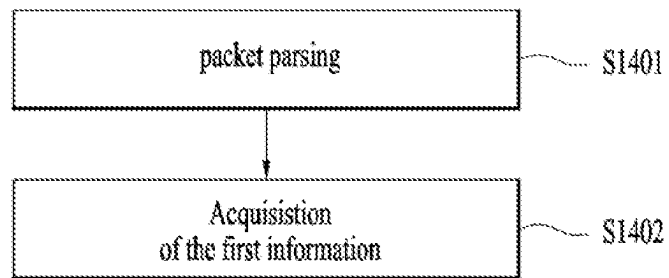


FIG. 15

1

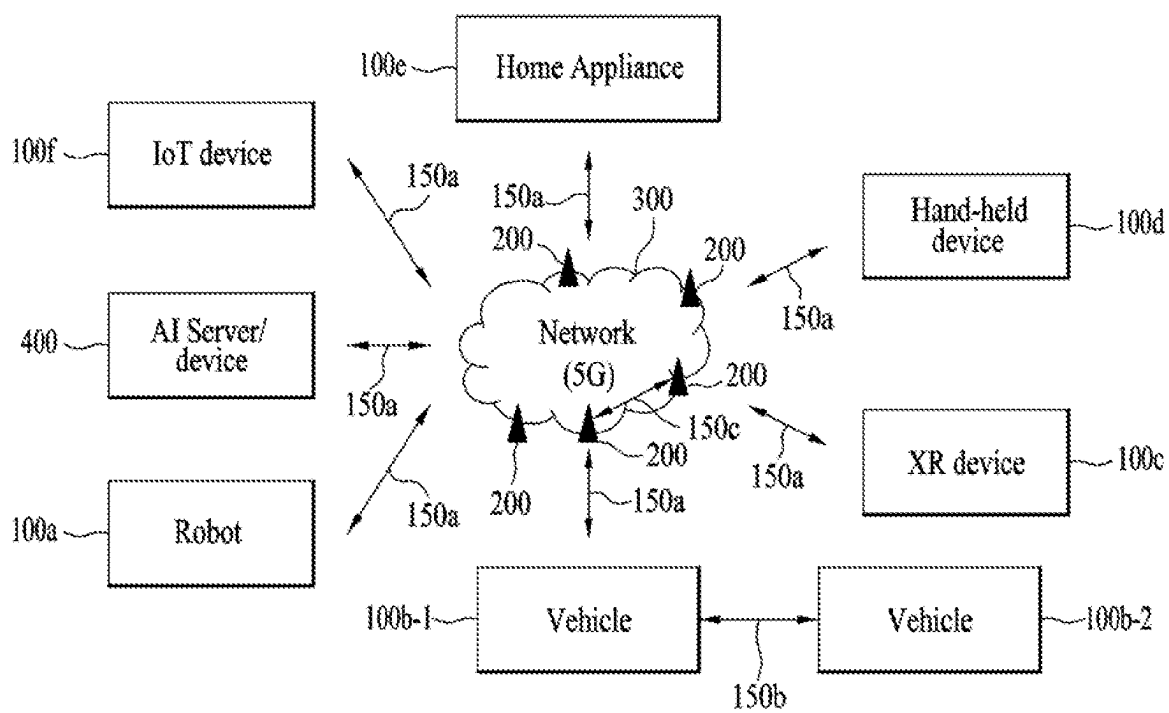


FIG. 16

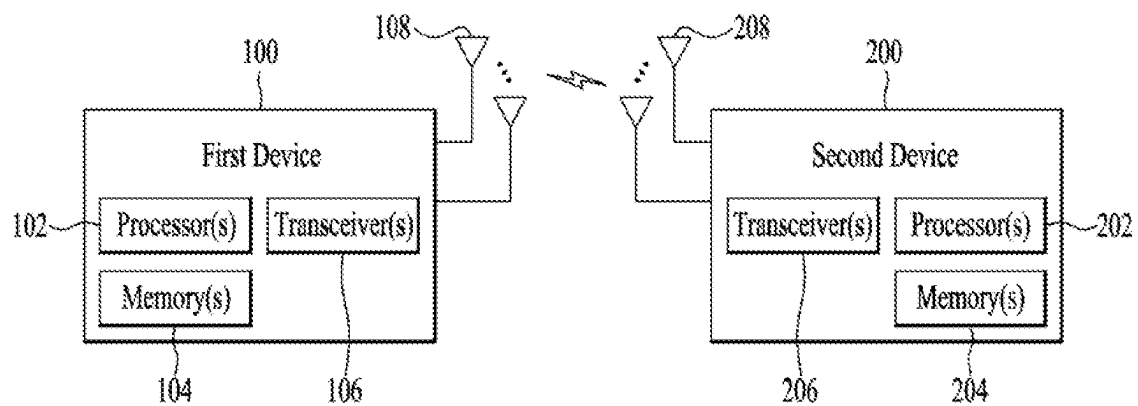


FIG. 17

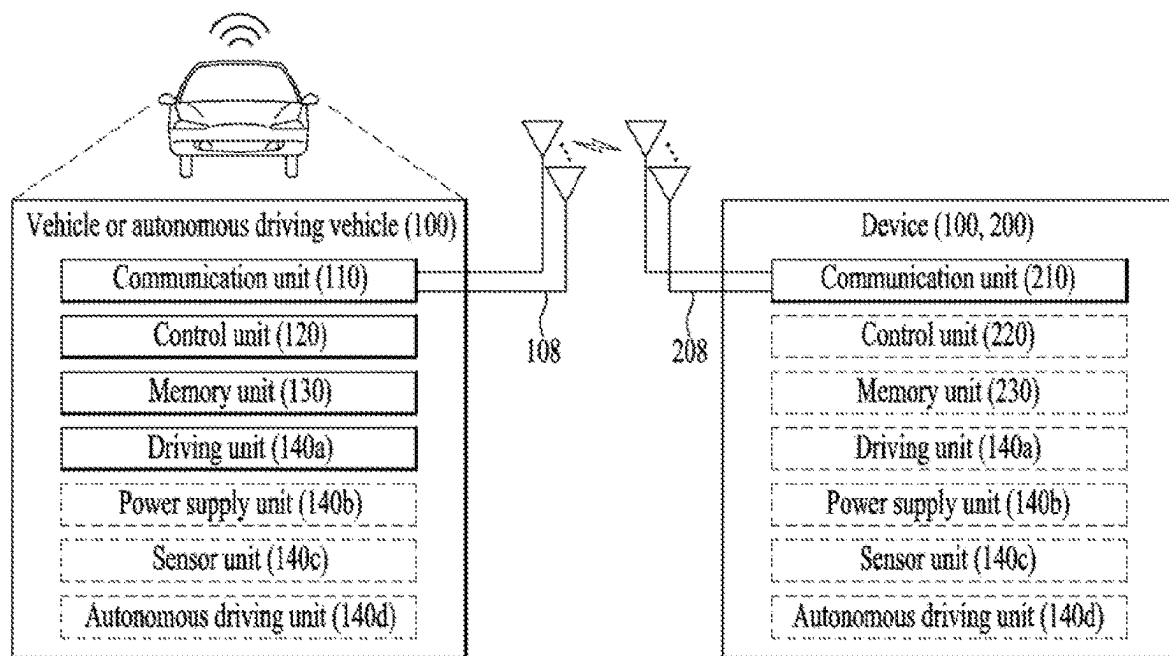


FIG. 18

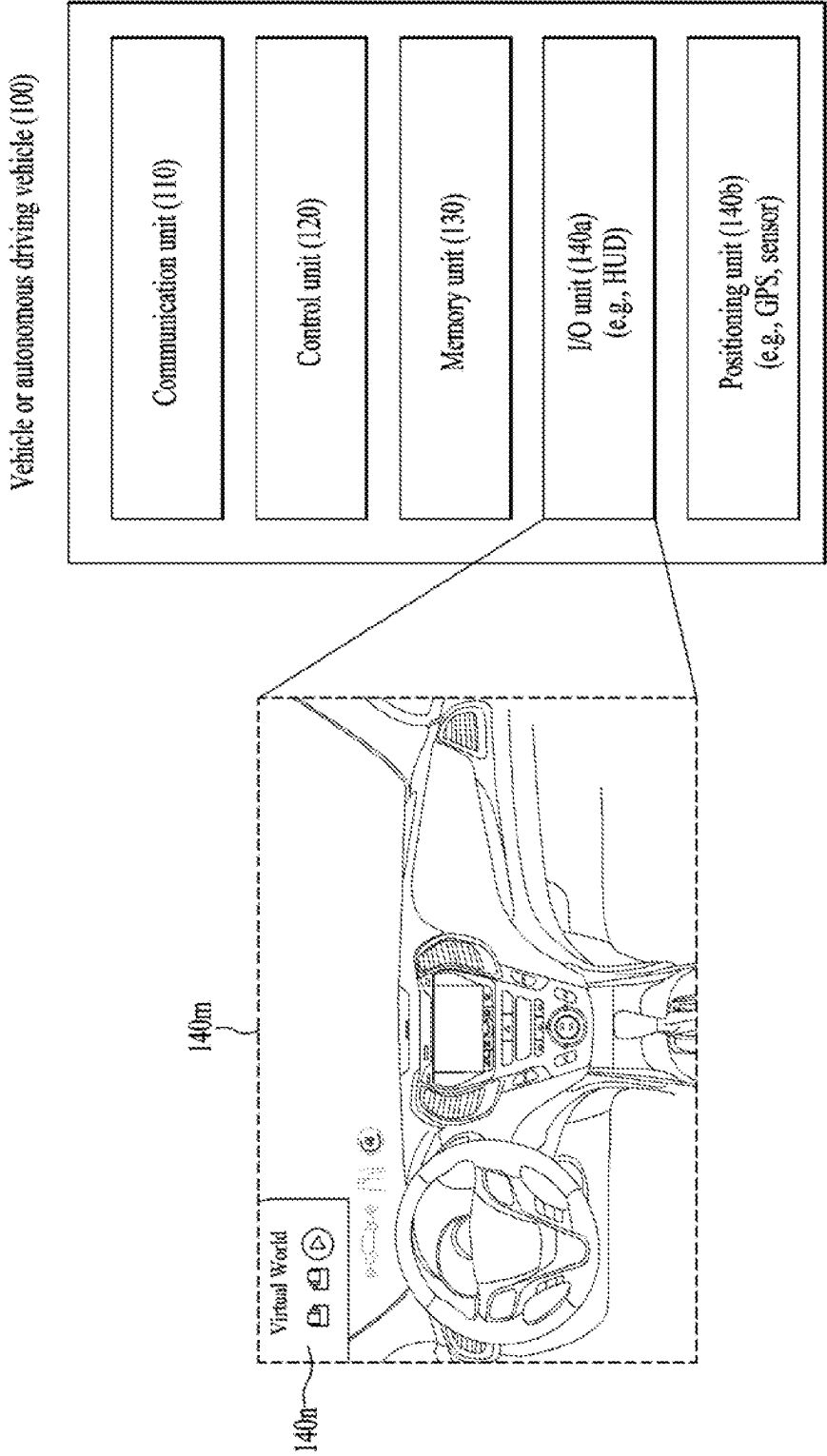


FIG. 19

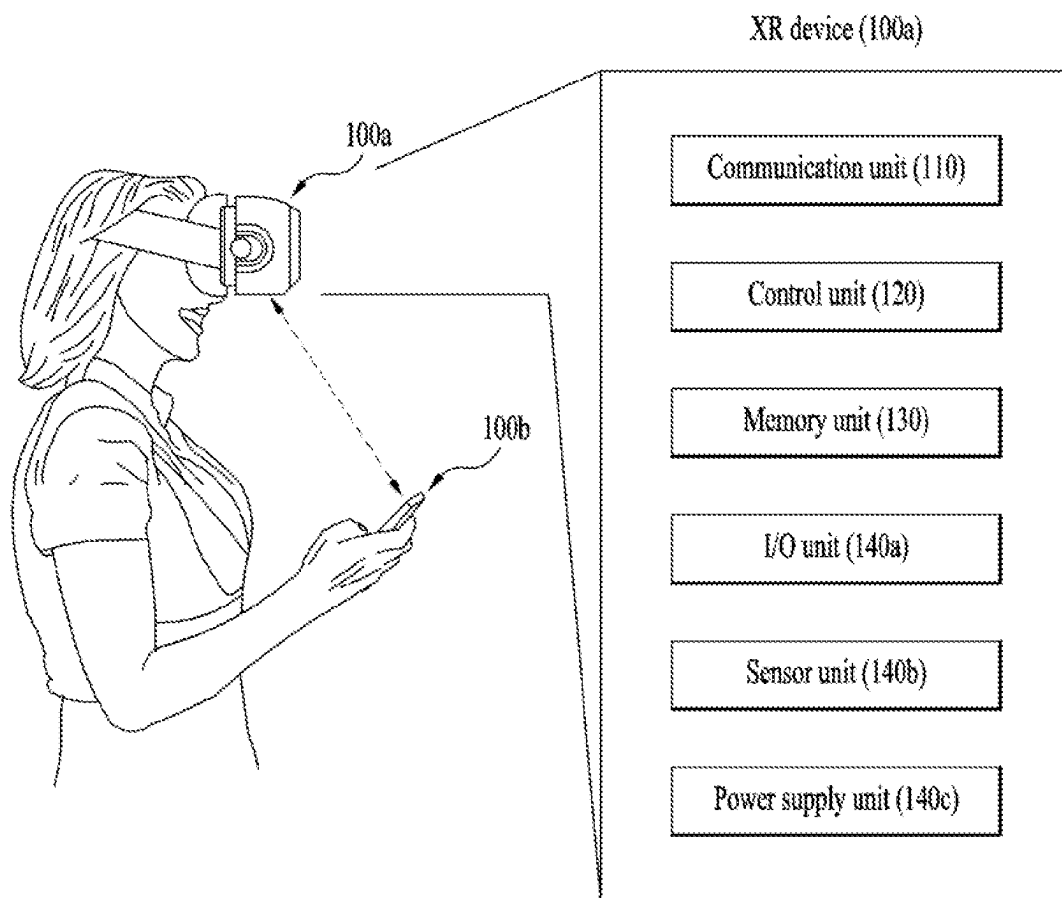


FIG. 20

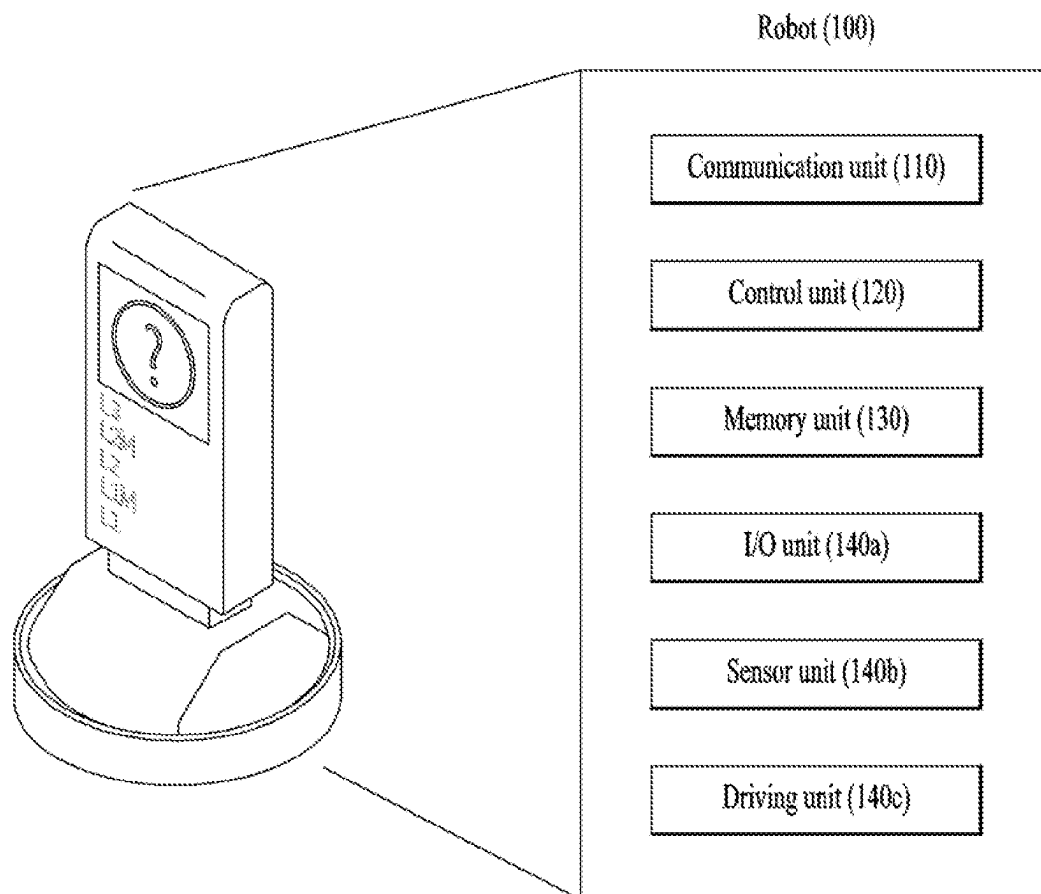
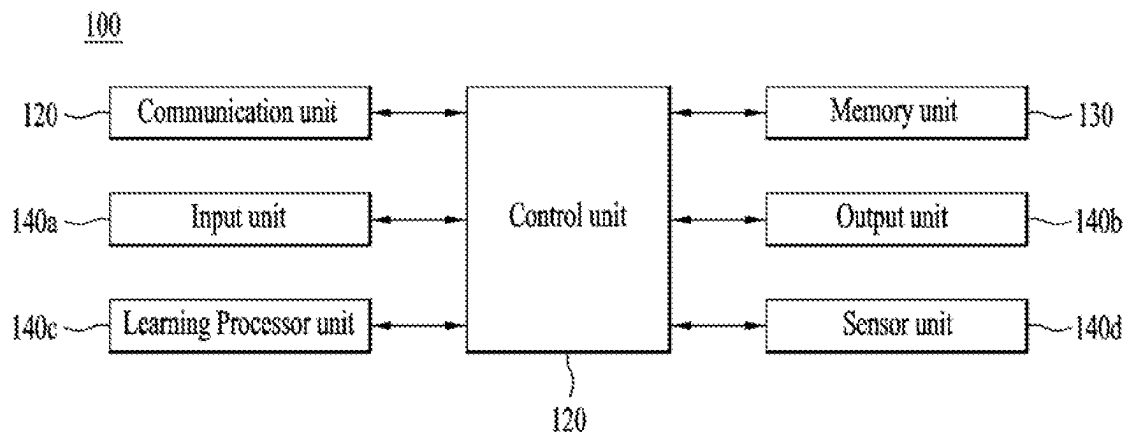


FIG. 21



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OPERATION METHOD OF UE RECEIVING V2X-RELATED MESSAGE IN WIRELESS COMMUNICATION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2021/002603, filed on Mar. 3, 2021, which claims the benefit of earlier filing date and right of priority to Korean Application No. 10-2020-0027670, filed on Mar. 5, 2020, the contents of which are all incorporated by reference herein in their entirety.

TECHNICAL FIELD

The following description relates to a wireless communication system and, more particularly, to an operation method of a sidelink UE related to processing/generation of an LTE-V2X PC5-related packet or an LTE-V2X Uu interface-related packet.

BACKGROUND ART

Wireless communication systems are being widely deployed to provide various types of communication services such as voice and data. In general, a wireless communication system is a multiple access system capable of supporting communication with multiple users by sharing available system resources (bandwidth, transmission power, etc.). Examples of the multiple access system include a code division multiple access (CDMA) system, a frequency division multiple access (FDMA) system, a time division multiple access (TDMA) system, an orthogonal frequency division multiple access (OFDMA) system, and a single carrier frequency division multiple access (SC-FDMA) system, and a multi carrier frequency division multiple access (MC-FDMA) system.

A wireless communication system uses various radio access technologies (RATs) such as long term evolution (LTE), LTE-advanced (LTE-A), and wireless fidelity (WiFi). 5th generation (5G) is such a wireless communication system. Three key requirement areas of 5G include (1) enhanced mobile broadband (eMBB), (2) massive machine type communication (mMTC), and (3) ultra-reliable and low latency communications (URLLC). Some use cases may require multiple dimensions for optimization, while others may focus only on one key performance indicator (KPI). 5G supports such diverse use cases in a flexible and reliable way.

eMBB goes far beyond basic mobile Internet access and covers rich interactive work, media and entertainment applications in the cloud or augmented reality (AR). Data is one of the key drivers for 5G and in the 5G era, we may for the first time see no dedicated voice service. In 5G, voice is expected to be handled as an application program, simply using data connectivity provided by a communication system. The main drivers for an increased traffic volume are the increase in the size of content and the number of applications requiring high data rates. Streaming services (audio and video), interactive video, and mobile Internet connectivity will continue to be used more broadly as more devices connect to the Internet. Many of these applications require always-on connectivity to push real time information and notifications to users. Cloud storage and applications are rapidly increasing for mobile communication platforms.

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This is applicable for both work and entertainment. Cloud storage is one particular use case driving the growth of uplink data rates. 5G will also be used for remote work in the cloud which, when done with tactile interfaces, requires much lower end-to-end latencies in order to maintain a good user experience. Entertainment, for example, cloud gaming and video streaming, is another key driver for the increasing need for mobile broadband capacity. Entertainment will be very essential on smart phones and tablets everywhere, including high mobility environments such as trains, cars and airplanes. Another use case is augmented reality (AR) for entertainment and information search, which requires very low latencies and significant instant data volumes.

One of the most expected 5G use cases is the functionality of actively connecting embedded sensors in every field, that is, mMTC. It is expected that there will be 20.4 billion potential Internet of things (IoT) devices by 2020. In industrial IoT, 5G is one of areas that play key roles in enabling smart city, asset tracking, smart utility, agriculture, and security infrastructure.

URLLC includes services which will transform industries with ultra-reliable/available, low latency links such as remote control of critical infrastructure and self-driving vehicles. The level of reliability and latency are vital to smart-grid control, industrial automation, robotics, drone control and coordination, and so on.

Now, multiple use cases will be described in detail.

5G may complement fiber-to-the home (FTTH) and cable-based broadband (or data-over-cable service interface specifications (DOCSIS)) as a means of providing streams at data rates of hundreds of megabits per second to giga bits per second. Such a high speed is required for TV broadcasts at or above a resolution of 4K (6K, 8K, and higher) as well as virtual reality (VR) and AR. VR and AR applications mostly include immersive sport games. A special network configuration may be required for a specific application program. For VR games, for example, game companies may have to integrate a core server with an edge network server of a network operator in order to minimize latency.

The automotive sector is expected to be a very important new driver for 5G, with many use cases for mobile communications for vehicles. For example, entertainment for passengers requires simultaneous high capacity and high mobility mobile broadband, because future users will expect to continue their good quality connection independent of their location and speed. Other use cases for the automotive sector are AR dashboards. These display overlay information on top of what a driver is seeing through the front window, identifying objects in the dark and telling the driver about the distances and movements of the objects. In the future, wireless modules will enable communication between vehicles themselves, information exchange between vehicles and supporting infrastructure and between vehicles and other connected devices (e.g., those carried by pedestrians). Safety systems may guide drivers on alternative courses of action to allow them to drive more safely and lower the risks of accidents. The next stage will be remote-controlled or self-driving vehicles. These require very reliable, very fast communication between different self-driving vehicles and between vehicles and infrastructure. In the future, self-driving vehicles will execute all driving activities, while drivers are focusing on traffic abnormality elusive to the vehicles themselves. The technical requirements for self-driving vehicles call for ultra-low latencies and ultra-high reliability, increasing traffic safety to levels humans cannot achieve.

Smart cities and smart homes, often referred to as smart society, will be embedded with dense wireless sensor networks. Distributed networks of intelligent sensors will identify conditions for cost- and energy-efficient maintenance of the city or home. A similar setup can be done for each home, where temperature sensors, window and heating controllers, burglar alarms, and home appliances are all connected wirelessly. Many of these sensors are typically characterized by low data rate, low power, and low cost, but for example, real time high definition (HD) video may be required in some types of devices for surveillance.

The consumption and distribution of energy, including heat or gas, is becoming highly decentralized, creating the need for automated control of a very distributed sensor network. A smart grid interconnects such sensors, using digital information and communications technology to gather and act on information. This information may include information about the behaviors of suppliers and consumers, allowing the smart grid to improve the efficiency, reliability, economics and sustainability of the production and distribution of fuels such as electricity in an automated fashion. A smart grid may be seen as another sensor network with low delays.

The health sector has many applications that may benefit from mobile communications. Communications systems enable telemedicine, which provides clinical health care at a distance. It helps eliminate distance barriers and may improve access to medical services that would often not be consistently available in distant rural communities. It is also used to save lives in critical care and emergency situations. Wireless sensor networks based on mobile communication may provide remote monitoring and sensors for parameters such as heart rate and blood pressure.

Wireless and mobile communications are becoming increasingly important for industrial applications. Wires are expensive to install and maintain, and the possibility of replacing cables with reconfigurable wireless links is a tempting opportunity for many industries. However, achieving this requires that the wireless connection works with a similar delay, reliability and capacity as cables and that its management is simplified. Low delays and very low error probabilities are new requirements that need to be addressed with 5G

Finally, logistics and freight tracking are important use cases for mobile communications that enable the tracking of inventory and packages wherever they are by using location-based information systems. The logistics and freight tracking use cases typically require lower data rates but need wide coverage and reliable location information.

A wireless communication system is a multiple access system that supports communication of multiple users by sharing available system resources (a bandwidth, transmission power, etc.). Examples of multiple access systems include a CDMA system, an FDMA system, a TDMA system, an OFDMA system, an SC-FDMA system, and an MC-FDMA system.

Sidelink (SL) refers to a communication scheme in which a direct link is established between user equipments (UEs) and the UEs directly exchange voice or data without intervention of a base station (BS). SL is considered as a solution of relieving the BS of the constraint of rapidly growing data traffic.

Vehicle-to-everything (V2X) is a communication technology in which a vehicle exchanges information with another vehicle, a pedestrian, and infrastructure by wired/wireless communication. V2X may be categorized into four types: vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I),

vehicle-to-network (V2N), and vehicle-to-pedestrian (V2P). V2X communication may be provided via a PC5 interface and/or a Uu interface.

As more and more communication devices demand larger communication capacities, there is a need for enhanced mobile broadband communication relative to existing RATs. Accordingly, a communication system is under discussion, for which services or UEs sensitive to reliability and latency are considered. The next-generation RAT in which eMBB, MTC, and URLLC are considered is referred to as new RAT or NR. In NR, V2X communication may also be supported.

FIG. 1 is a diagram illustrating V2X communication based on pre-NR RAT and V2X communication based on NR in comparison.

For V2X communication, a technique of providing safety service based on V2X messages such as basic safety message (BSM), cooperative awareness message (CAM), and decentralized environmental notification message (DENM) was mainly discussed in the pre-NR RAT. The V2X message may include location information, dynamic information, and attribute information. For example, a UE may transmit a CAM of a periodic message type and/or a DENM of an event-triggered type to another UE.

For example, the CAM may include basic vehicle information including dynamic state information such as a direction and a speed, vehicle static data such as dimensions, an external lighting state, path details, and so on. For example, the UE may broadcast the CAM which may have a latency less than 100 ms. For example, when an unexpected incident occurs, such as breakage or an accident of a vehicle, the UE may generate the DENM and transmit the DENM to another UE. For example, all vehicles within the transmission range of the UE may receive the CAM and/or the DENM. In this case, the DENM may have priority over the CAM.

In relation to V2X communication, various V2X scenarios are presented in NR. For example, the V2X scenarios include vehicle platooning, advanced driving, extended sensors, and remote driving.

For example, vehicles may be dynamically grouped and travel together based on vehicle platooning. For example, to perform platoon operations based on vehicle platooning, the vehicles of the group may receive periodic data from a leading vehicle. For example, the vehicles of the group may widen or narrow their gaps based on the periodic data.

For example, a vehicle may be semi-automated or full-automated based on advanced driving. For example, each vehicle may adjust a trajectory or maneuvering based on data obtained from a nearby vehicle and/or a nearby logical entity. For example, each vehicle may also share a dividing intention with nearby vehicles.

Based on extended sensors, for example, raw or processed data obtained through local sensor or live video data may be exchanged between vehicles, logical entities, terminals of pedestrians and/or V2X application servers. Accordingly, a vehicle may perceive an advanced environment relative to an environment perceivable by its sensor.

Based on remote driving, for example, a remote driver or a V2X application may operate or control a remote vehicle on behalf of a person incapable of driving or in a dangerous environment. For example, when a path may be predicted as in public transportation, cloud computing-based driving may be used in operating or controlling the remote vehicle. For example, access to a cloud-based back-end service platform may also be used for remote driving.

A scheme of specifying service requirements for various V2X scenarios including vehicle platooning, advanced driv-

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ing, extended sensors, and remote driving is under discussion in NR-based V2X communication.

DISCLOSURE

Technical Problem

Embodiment(s) are directed to an operation method related to processing/generation of an LTE-V2X PC5-related packet or an LTE-V2X Uu interface-related packet.

Technical Solution

According to an embodiment, provided herein is an operation method of a first user equipment (UE) in a wireless communication system, including receiving, by the first UE, a vehicle-to-everything (V2X)-related packet from one of a second UE and a base station; and acquiring, by the first UE, a V2X-related message based on parsing of the V2X-related packet. The V2X-related packet includes header information indicating whether the V2X-related packet is a long-term evolution (LTE)-V2X PC5-related packet or an LTE-V2X Uu interface-related packet.

According to another embodiment, provided herein is a first user equipment (UE) in a wireless communication system, including at least one processor; and at least one computer memory operably connectable to the at least one processor and storing instructions that, when executed, cause the at least one processor to perform operations. The operations include receiving a vehicle-to-everything (V2X)-related packet from one of a second UE and a base station; and acquiring a V2X-related message based on parsing of the V2X-related packet. The V2X-related packet includes header information indicating whether the V2X-related packet is a long-term evolution (LTE)-V2X PC5-related packet or an LTE-V2X Uu interface-related packet.

According to another embodiment, provided herein is a processor for performing operations for a first user equipment (UE) in a wireless communication system. The operations include receiving a vehicle-to-everything (V2X)-related packet from one of a second UE and a base station; and acquiring a V2X-related message based on parsing of the V2X-related packet. The V2X-related packet includes header information indicating whether the V2X-related packet is a long-term evolution (LTE)-V2X PC5-related packet or an LTE-V2X Uu interface-related packet.

According to another embodiment, provided herein is a non-volatile computer readable storage medium storing at least one computer program including instructions that, when executed by at least one processor, cause the at least one processor to perform operations for a user equipment (UE). The operations include receiving a vehicle-to-everything (V2X)-related packet from one of a second UE and a base station; and acquiring a V2X-related message based on parsing of the V2X-related packet. The V2X-related packet includes header information indicating whether the V2X-related packet is a long-term evolution (LTE)-V2X PC5-related packet or an LTE-V2X Uu interface-related packet.

A network layer of the first UE may determine whether to process the V2X-related packet through a basic transport protocol (BTP) or through a transmission control protocol (TCP), based on the header information.

The V2X-related packet may be transmitted to a higher layer through a GeoNetworking protocol.

The header information may be a traffic class field of a GeoNetworking common header.

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An integer, itsGnIfType, defining an interface type of the GeoNetworking protocol may indicate which one of an LTE-V2X PC5 interface or an LTE-V2X Uu-interface the interface type is.

5 An ASN.1 module related to the integer itsGnIfType may indicate which one of the LTE-V2X PC5 interface and the LTE-V2X Uu-interface the interface type is.

The V2X-related packet may be generated based on a communication profile of GN-DATA.request indicating which one of the LTE-V2X PC5 interface and the LTE-V2X Uu interface the interface type is.

The V2X-related packet may be generated based on a GN-DATA.request service primitive including a range parameter and a DirectCommunication parameter.

15 The range parameter may indicate one of a short range and a long range, and the DirectCommunication parameter may indicate one of direct communication and indirect communication.

20 The V2X-related packet may be an intelligent transport systems (ITS) packet.

Advantageous Effects

According to an embodiment, a problem that a GeoNetworking protocol incorrectly transmits a TCP/IP packet used for LTE-V2X Uu through a BTP may be solved.

BRIEF DESCRIPTION OF DRAWINGS

30 The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this application, illustrate embodiment(s) of the disclosure and together with the description serve to explain the principle of the disclosure. In the drawings:

FIG. 1 is a diagram comparing vehicle-to-everything (V2X) communication based on pre-new radio access technology (pre-NR) with V2X communication based on NR;

40 FIG. 2 is a diagram illustrating the structure of a long term evolution (LTE) system according to an embodiment of the present disclosure;

FIG. 3 is a diagram illustrating user-plane and control-plane radio protocol architectures according to an embodiment of the present disclosure;

45 FIG. 4 is a diagram illustrating the structure of an NR system according to an embodiment of the present disclosure;

FIG. 5 is a diagram illustrating functional split between a next generation radio access network (NG-RAN) and a 5th generation core network (5GC) according to an embodiment of the present disclosure;

FIG. 6 is a diagram illustrating the structure of an NR radio frame to which embodiment(s) of the present disclosure is applicable;

55 FIG. 7 is a diagram illustrating a slot structure of an NR frame according to an embodiment of the present disclosure;

FIG. 8 is a diagram illustrating radio protocol architectures for sidelink (SL) communication according to an embodiment of the present disclosure;

60 FIG. 9 is a diagram illustrating radio protocol architectures for SL communication according to an embodiment of the present disclosure;

FIG. 10 illustrates a procedure in which a UE performs V2X or SL communication according to a transmission mode according to an embodiment of the present disclosure;

FIGS. 11 to 14 are diagrams for explaining embodiment(s); and

FIGS. 15 to 21 are diagrams for explaining various devices to which embodiment(s) are applicable.

BEST MODE

In various embodiments of the present disclosure, “I” and “,” should be interpreted as “and/or”. For example, “A/B” may mean “A and/or B”. Further, “A, B” may mean “A and/or B”. Further, “A/B/C” may mean “at least one of A, B and/or C”. Further, “A, B, C” may mean “at least one of A, B and/or C”.

In various embodiments of the present disclosure, “or” should be interpreted as “and/or”. For example, “A or B” may include “only A”, “only B”, and/or “both A and B”. In other words, “or” should be interpreted as “additionally or alternatively”.

Techniques described herein may be used in various wireless access systems such as code division multiple access (CDMA), frequency division multiple access (FDMA), time division multiple access (TDMA), orthogonal frequency division multiple access (OFDMA), single carrier-frequency division multiple access (SC-FDMA), and so on. CDMA may be implemented as a radio technology such as universal terrestrial radio access (UTRA) or CDMA2000. TDMA may be implemented as a radio technology such as global system for mobile communications (GSM)/general packet radio service (GPRS)/Enhanced Data Rates for GSM Evolution (EDGE). OFDMA may be implemented as a radio technology such as IEEE 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, evolved-UTRA (E-UTRA), or the like. IEEE 802.16m is an evolution of IEEE 802.16e, offering backward compatibility with an IRRR 802.16e-based system. UTRA is a part of universal mobile telecommunications system (UMTS). 3rd generation partnership project (3GPP) long term evolution (LTE) is a part of evolved UMTS (E-UMTS) using evolved UTRA (E-UTRA). 3GPP LTE employs OFDMA for downlink (DL) and SC-FDMA for uplink (UL). LTE-advanced (LTE-A) is an evolution of 3GPP LTE.

A successor to LTE-A, 5th generation (5G) new radio access technology (NR) is a new clean-state mobile communication system characterized by high performance, low latency, and high availability. 5G NR may use all available spectral resources including a low frequency band below 1 GHz, an intermediate frequency band between 1 GHz and 10 GHz, and a high frequency (millimeter) band of 24 GHz or above.

While the following description is given mainly in the context of LTE-A or 5G NR for the clarity of description, the technical idea of an embodiment of the present disclosure is not limited thereto.

FIG. 2 illustrates the structure of an LTE system according to an embodiment of the present disclosure. This may also be called an evolved UMTS terrestrial radio access network (E-UTRAN) or LTE/LTE-A system.

Referring to FIG. 2, the E-UTRAN includes evolved Node Bs (eNBs) 20 which provide a control plane and a user plane to UEs 10. A UE 10 may be fixed or mobile, and may also be referred to as a mobile station (MS), user terminal (UT), subscriber station (SS), mobile terminal (MT), or wireless device. An eNB 20 is a fixed station communication with the UE 10 and may also be referred to as a base station (BS), a base transceiver system (BTS), or an access point.

eNBs 20 may be connected to each other via an X2 interface. An eNB 20 is connected to an evolved packet core (EPC) 30 via an S1 interface. More specifically, the eNB 20

is connected to a mobility management entity (MME) via an S1-MME interface and to a serving gateway (S-GW) via an S1-U interface.

The EPC 30 includes an MME, an S-GW, and a packet data network-gateway (P-GW). The MME has access information or capability information about UEs, which are mainly used for mobility management of the UEs. The S-GW is a gateway having the E-UTRAN as an end point, and the P-GW is a gateway having a packet data network (PDN) as an end point.

Based on the lowest three layers of the open system interconnection (OSI) reference model known in communication systems, the radio protocol stack between a UE and a network may be divided into Layer 1 (L1), Layer 2 (L2) and Layer 3 (L3). These layers are defined in pairs between a UE and an Evolved UTRAN (E-UTRAN), for data transmission via the Uu interface. The physical (PHY) layer at L1 provides an information transfer service on physical channels. The radio resource control (RRC) layer at L3 functions to control radio resources between the UE and the network. For this purpose, the RRC layer exchanges RRC messages between the UE and an eNB.

FIG. 3(a) illustrates a user-plane radio protocol architecture according to an embodiment of the disclosure.

FIG. 3(b) illustrates a control-plane radio protocol architecture according to an embodiment of the disclosure. A user plane is a protocol stack for user data transmission, and a control plane is a protocol stack for control signal transmission.

Referring to FIGS. 3(a) and 3(b), the PHY layer provides an information transfer service to its higher layer on physical channels. The PHY layer is connected to the medium access control (MAC) layer through transport channels and data is transferred between the MAC layer and the PHY layer on the transport channels. The transport channels are divided according to features with which data is transmitted via a radio interface.

Data is transmitted on physical channels between different PHY layers, that is, the PHY layers of a transmitter and a receiver. The physical channels may be modulated in orthogonal frequency division multiplexing (OFDM) and use time and frequencies as radio resources.

The MAC layer provides services to a higher layer, radio link control (RLC) on logical channels. The MAC layer provides a function of mapping from a plurality of logical channels to a plurality of transport channels. Further, the MAC layer provides a logical channel multiplexing function by mapping a plurality of logical channels to a single transport channel. A MAC sublayer provides a data transmission service on the logical channels.

The RLC layer performs concatenation, segmentation, and reassembly for RLC serving data units (SDUs). In order to guarantee various quality of service (QoS) requirements of each radio bearer (RB), the RLC layer provides three operation modes, transparent mode (TM), unacknowledged mode (UM), and acknowledged Mode (AM). An AM RLC provides error correction through automatic repeat request (ARQ).

The RRC layer is defined only in the control plane and controls logical channels, transport channels, and physical channels in relation to configuration, reconfiguration, and release of RBs. An RB refers to a logical path provided by L1 (the PHY layer) and L2 (the MAC layer, the RLC layer, and the packet data convergence protocol (PDCP) layer), for data transmission between the UE and the network.

The user-plane functions of the PDCP layer include user data transmission, header compression, and ciphering. The

control-plane functions of the PDCP layer include control-plane data transmission and ciphering/integrity protection.

RB establishment amounts to a process of defining radio protocol layers and channel features and configuring specific parameters and operation methods in order to provide a specific service. RBs may be classified into two types, signaling radio bearer (SRB) and data radio bearer (DRB). The SRB is used as a path in which an RRC message is transmitted on the control plane, whereas the DRB is used as a path in which user data is transmitted on the user plane.

Once an RRC connection is established between the RRC layer of the UE and the RRC layer of the E-UTRAN, the UE is placed in RRC_CONNECTED state, and otherwise, the UE is placed in RRC_IDLE state. In NR, RRC_INACTIVE state is additionally defined. A UE in the RRC_INACTIVE state may maintain a connection to a core network, while releasing a connection from an eNB.

DL transport channels carrying data from the network to the UE include a broadcast channel (BCH) on which system information is transmitted and a DL shared channel (DL SCH) on which user traffic or a control message is transmitted. Traffic or a control message of a DL multicast or broadcast service may be transmitted on the DL-SCH or a DL multicast channel (DL MCH). UL transport channels carrying data from the UE to the network include a random access channel (RACH) on which an initial control message is transmitted and an UL shared channel (UL SCH) on which user traffic or a control message is transmitted.

The logical channels which are above and mapped to the transport channels include a broadcast control channel (BCCH), a paging control channel (PCCH), a common control channel (CCCH), a multicast control channel (MCCH), and a multicast traffic channel (MTCH).

A physical channel includes a plurality of OFDM symbol in the time domain by a plurality of subcarriers in the frequency domain. One subframe includes a plurality of OFDM symbols in the time domain. An RB is a resource allocation unit defined by a plurality of OFDM symbols by a plurality of subcarriers. Further, each subframe may use specific subcarriers of specific OFDM symbols (e.g., the first OFDM symbol) in a corresponding subframe for a physical DL control channel (PDCCH), that is, an L1/L2 control channel. A transmission time interval (TTI) is a unit time for subframe transmission.

FIG. 4 illustrates the structure of an NR system according to an embodiment of the present disclosure.

Referring to FIG. 4, a next generation radio access network (NG-RAN) may include a next generation Node B (gNB) and/or an eNB, which provides user-plane and control-plane protocol termination to a UE. In FIG. 4, the NG-RAN is shown as including only gNBs, by way of example. A gNB and an eNB are connected to each other via an Xn interface. The gNB and the eNB are connected to a 5G core network (5GC) via an NG interface. More specifically, the gNB and the eNB are connected to an access and mobility management function (AMF) via an NG-C interface and to a user plane function (UPF) via an NG-U interface.

FIG. 5 illustrates functional split between the NG-RAN and the 5GC according to an embodiment of the present disclosure.

Referring to FIG. 5, a gNB may provide functions including inter-cell radio resource management (RRM), radio admission control, measurement configuration and provision, and dynamic resource allocation. The AMF may provide functions such as non-access stratum (NAS) security and idle-state mobility processing. The UPF may provide

functions including mobility anchoring and protocol data unit (PDU) processing. A session management function (SMF) may provide functions including UE Internet protocol (IP) address allocation and PDU session control.

FIG. 6 illustrates a radio frame structure in NR, to which embodiment(s) of the present disclosure is applicable.

Referring to FIG. 6, a radio frame may be used for UL transmission and DL transmission in NR. A radio frame is 10 ms in length, and may be defined by two 5-ms half-frames. An HF may include five 1-ms subframes. A subframe may be divided into one or more slots, and the number of slots in an SF may be determined according to a subcarrier spacing (SCS). Each slot may include 12 or 14 OFDM(A) symbols according to a cyclic prefix (CP).

In a normal CP (NCP) case, each slot may include 14 symbols, whereas in an extended CP (ECP) case, each slot may include 12 symbols. Herein, a symbol may be an OFDM symbol (or CP-OFDM symbol) or an SC-FDMA symbol (or DFT-s-OFDM symbol).

Table 1 below lists the number of symbols per slot N_{symbol}^{slot} , the number of slots per frame $N_{slot}^{frame, u}$, and the number of slots per subframe $N_{slot}^{subframe, u}$ according to an SCS configuration μ in the NCP case.

TABLE 1

SCS (15*2 ^{μ} u)	N_{symbol}^{slot}	$N_{slot}^{frame, u}$	$N_{slot}^{subframe, u}$
15 KHz (u = 0)	14	10	1
30 KHz (u = 1)	14	20	2
60 KHz (u = 2)	14	40	4
120 KHz (u = 3)	14	80	8
240 KHz (u = 4)	14	160	16

Table 2 below lists the number of symbols per slot, the number of slots per frame, and the number of slots per subframe according to an SCS in the ECP case.

TABLE 2

SCS (15*2 ^{μ} u)	N_{symbol}^{slot}	$N_{slot}^{frame, u}$	$N_{slot}^{subframe, u}$
60 KHz (u = 2)	12	40	4

In the NR system, different OFDM(A) numerologies (e.g., SCSs, CP lengths, and so on) may be configured for a plurality of cells aggregated for one UE. Accordingly, the (absolute time) duration of a time resource including the same number of symbols (e.g., a subframe, slot, or TTI) (collectively referred to as a time unit (TU) for convenience) may be configured to be different for the aggregated cells.

In NR, various numerologies or SCSs may be supported to support various 5G services. For example, with an SCS of 15 kHz, a wide area in traditional cellular bands may be supported, while with an SCS of 30 kHz/60 kHz, a dense urban area, a lower latency, and a wide carrier bandwidth may be supported. With an SCS of 60 kHz or higher, a bandwidth larger than 24.25 GHz may be supported to overcome phase noise.

An NR frequency band may be defined by two types of frequency ranges, FR1 and FR2. The numerals in each frequency range may be changed. For example, the two types of frequency ranges may be given in Table 3. In the NR system, FR1 may be a "sub 6 GHz range" and FR2 may be an "above 6 GHz range" called millimeter wave (mmW).

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TABLE 3

Frequency Range designation	Corresponding frequency range	Subcarrier Spacing (SCS)
FR1	450 MHz-6000 MHz	15, 30, 60 kHz
FR2	24250 MHz-52600 MHz	60, 120, 240 kHz

As mentioned above, the numerals in a frequency range may be changed in the NR system. For example, FR1 may range from 410 MHz to 7125 MHz as listed in Table 4. That is, FR1 may include a frequency band of 6 GHz (or 5850, 5900, and 5925 MHz) or above. For example, the frequency band of 6 GHz (or 5850, 5900, and 5925 MHz) or above may include an unlicensed band. The unlicensed band may be used for various purposes, for example, vehicle communication (e.g., autonomous driving).

TABLE 4

Frequency Range designation	Corresponding frequency range	Subcarrier Spacing (SCS)
FR1	410 MHz-7125 MHz	15, 30, 60 kHz
FR2	24250 MHz-52600 MHz	60, 120, 240 kHz

FIG. 7 illustrates a slot structure in an NR frame according to an embodiment of the present disclosure.

Referring to FIG. 7, a slot includes a plurality of symbols in the time domain. For example, one slot may include 14 symbols in an NCP case and 12 symbols in an ECP case. Alternatively, one slot may include 7 symbols in an NCP case and 6 symbols in an ECP case.

A carrier includes a plurality of subcarriers in the frequency domain. An RB may be defined by a plurality of (e.g., 12) consecutive subcarriers in the frequency domain. A bandwidth part (BWP) may be defined by a plurality of consecutive (physical) RBs ((P)RBs) in the frequency domain and correspond to one numerology (e.g., SCS, CP length, or the like). A carrier may include up to N (e.g., 5) BWPs. Data communication may be conducted in an activated BWP. Each element may be referred to as a resource element (RE) in a resource grid, to which one complex symbol may be mapped.

A radio interface between UEs or a radio interface between a UE and a network may include L1, L2, and L3. In various embodiments of the present disclosure, L1 may refer to the PHY layer. For example, L2 may refer to at least one of the MAC layer, the RLC layer, the PDCH layer, or the SDAP layer. For example, L3 may refer to the RRC layer.

Now, a description will be given of sidelink (SL) communication.

FIG. 8 illustrates a radio protocol architecture for SL communication according to an embodiment of the present disclosure. Specifically, FIG. 8(a) illustrates a user-plane protocol stack in LTE, and FIG. 8(b) illustrates a control-plane protocol stack in LTE.

FIG. 9 illustrates a radio protocol architecture for SL communication according to an embodiment of the present disclosure. Specifically, FIG. 9(a) illustrates a user-plane protocol stack in NR, and FIG. 9(b) illustrates a control-plane protocol stack in NR.

Resource allocation in SL will be described below.

FIG. 10 illustrates a procedure of performing V2X or SL communication according to a transmission mode in a UE according to an embodiment of the present disclosure. In various embodiments of the present disclosure, a transmission mode may also be referred to as a mode or a resource

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allocation mode. For the convenience of description, a transmission mode in LTE may be referred to as an LTE transmission mode, and a transmission mode in NR may be referred to as an NR resource allocation mode.

For example, FIG. 10(a) illustrates a UE operation related to LTE transmission mode 1 or LTE transmission mode 3. Alternatively, for example, FIG. 10(a) illustrates a UE operation related to NR resource allocation mode 1. For example, LTE transmission mode 1 may be applied to general SL communication, and LTE transmission mode 3 may be applied to V2X communication.

For example, FIG. 10(b) illustrates a UE operation related to LTE transmission mode 2 or LTE transmission mode 4. Alternatively, for example, FIG. 10(b) illustrates a UE operation related to NR resource allocation mode 2.

Referring to FIG. 10(a), in LTE transmission mode 1, LTE transmission mode 3, or NR resource allocation mode 1, a BS may schedule SL resources to be used for SL transmission of a UE. For example, the BS may perform resource scheduling for UE1 through a PDCCH (more specifically, DL control information (DCI)), and UE1 may perform V2X or SL communication with UE2 according to the resource scheduling. For example, UE1 may transmit sidelink control information (SCI) to UE2 on a PSCCH, and then transmit data based on the SCI to UE2 on a PSSCH.

For example, in NR resource allocation mode 1, a UE may be provided with or allocated resources for one or more SL transmissions of one transport block (TB) by a dynamic grant from the BS. For example, the BS may provide the UE with resources for transmission of a PSCCH and/or a PSSCH by the dynamic grant. For example, a transmitting UE may report an SL hybrid automatic repeat request (SL HARQ) feedback received from a receiving UE to the BS. In this case, PUCCH resources and a timing for reporting the SL HARQ feedback to the BS may be determined based on an indication in a PDCCH, by which the BS allocates resources for SL transmission.

For example, the DCI may indicate a slot offset between the DCI reception and a first SL transmission scheduled by the DCI. For example, a minimum gap between the DCI that schedules the SL transmission resources and the resources of the first scheduled SL transmission may not be smaller than a processing time of the UE.

For example, in NR resource allocation mode 1, the UE may be periodically provided with or allocated a resource set for a plurality of SL transmissions through a configured grant from the BS. For example, the grant to be configured may include configured grant type 1 or configured grant type 2. For example, the UE may determine a TB to be transmitted in each occasion indicated by a given configured grant.

For example, the BS may allocate SL resources to the UE in the same carrier or different carriers.

For example, an NR gNB may control LTE-based SL communication. For example, the NR gNB may transmit NR DCI to the UE to schedule LTE SL resources. In this case, for example, a new RNTI may be defined to scramble the NR DCI. For example, the UE may include an NR SL module and an LTE SL module.

For example, after the UE including the NR SL module and the LTE SL module receives NR SL DCI from the gNB, the NR SL module may convert the NR SL DCI into LTE DCI type 5A, and transmit LTE DCI type 5A to the LTE SL module every Xms. For example, after the LTE SL module receives LTE DCI format 5A from the NR SL module, the LTE SL module may activate and/or release a first LTE subframe after Z ms. For example, X may be dynamically

indicated by a field of the DCI. For example, a minimum value of X may be different according to a UE capability. For example, the UE may report a single value according to its UE capability. For example, X may be positive.

Referring to FIG. 10(b), in LTE transmission mode 2, LTE transmission mode 4, or NR resource allocation mode 2, the UE may determine SL transmission resources from among SL resources preconfigured or configured by the BS/network. For example, the preconfigured or configured SL resources may be a resource pool. For example, the UE may autonomously select or schedule SL transmission resources. For example, the UE may select resources in a configured resource pool on its own and perform SL communication in the selected resources. For example, the UE may select resources within a selection window on its own by a sensing and resource (re)selection procedure. For example, the sensing may be performed on a subchannel basis. UE1, which has autonomously selected resources in a resource pool, may transmit SCI to UE2 on a PSSCH and then transmit data based on the SCI to UE2 on a PSSCH.

For example, a UE may help another UE with SL resource selection. For example, in NR resource allocation mode 2, the UE may be configured with a grant configured for SL transmission. For example, in NR resource allocation mode 2, the UE may schedule SL transmission for another UE. For example, in NR resource allocation mode 2, the UE may reserve SL resources for blind retransmission.

For example, in NR resource allocation mode 2, UE1 may indicate the priority of SL transmission to UE2 by SCI. For example, UE2 may decode the SCI and perform sensing and/or resource (re)selection based on the priority. For example, the resource (re)selection procedure may include identifying candidate resources in a resource selection window by UE2 and selecting resources for (re)transmission from among the identified candidate resources by UE2. For example, the resource selection window may be a time interval during which the UE selects resources for SL transmission. For example, after UE2 triggers resource (re)selection, the resource selection window may start at $T1 \geq 0$, and may be limited by the remaining packet delay budget of UE2. For example, when specific resources are indicated by the SCI received from UE1 by the second UE and an L1 SL reference signal received power (RSRP) measurement of the specific resources exceeds an SL RSRP threshold in the step of identifying candidate resources in the resource selection window by UE2, UE2 may not determine the specific resources as candidate resources. For example, the SL RSRP threshold may be determined based on the priority of SL transmission indicated by the SCI received from UE1 by UE2 and the priority of SL transmission in the resources selected by UE2.

For example, the L1 SL RSRP may be measured based on an SL demodulation reference signal (DMRS). For example, one or more PSSCH DMRS patterns may be configured or preconfigured in the time domain for each resource pool. For example, PDSCH DMRS configuration type 1 and/or type 2 may be identical or similar to a PSSCH DMRS pattern in the frequency domain. For example, an accurate DMRS pattern may be indicated by the SCI. For example, in NR resource allocation mode 2, the transmitting UE may select a specific DMRS pattern from among DMRS patterns configured or preconfigured for the resource pool.

For example, in NR resource allocation mode 2, the transmitting UE may perform initial transmission of a TB without reservation based on the sensing and resource (re)selection procedure. For example, the transmitting UE may reserve SL resources for initial transmission of a second

TB using SCI associated with a first TB based on the sensing and resource (re)selection procedure.

For example, in NR resource allocation mode 2, the UE may reserve resources for feedback-based PSSCH retransmission through signaling related to a previous transmission of the same TB. For example, the maximum number of SL resources reserved for one transmission, including a current transmission, may be 2, 3 or 4. For example, the maximum number of SL resources may be the same regardless of whether HARQ feedback is enabled. For example, the maximum number of HARQ (re)transmissions for one TB may be limited by a configuration or preconfiguration. For example, the maximum number of HARQ (re)transmissions may be up to 32. For example, if there is no configuration or preconfiguration, the maximum number of HARQ (re)transmissions may not be specified. For example, the configuration or preconfiguration may be for the transmitting UE. For example, in NR resource allocation mode 2, HARQ feedback for releasing resources which are not used by the UE may be supported.

For example, in NR resource allocation mode 2, the UE may indicate one or more subchannels and/or slots used by the UE to another UE by SCI. For example, the UE may indicate one or more subchannels and/or slots reserved for PSSCH (re)transmission by the UE to another UE by SCI. For example, a minimum allocation unit of SL resources may be a slot. For example, the size of a subchannel may be configured or preconfigured for the UE.

SCI will be described below.

While control information transmitted from a BS to a UE on a PDCCH is referred to as DCI, control information transmitted from one UE to another UE on a PSSCH may be referred to as SCI. For example, the UE may know the starting symbol of the PSSCH and/or the number of symbols in the PSSCH before decoding the PSSCH. For example, the SCI may include SL scheduling information. For example, the UE may transmit at least one SCI to another UE to schedule the PSSCH. For example, one or more SCI formats may be defined.

For example, the transmitting UE may transmit the SCI to the receiving UE on the PSSCH. The receiving UE may decode one SCI to receive the PSSCH from the transmitting UE.

For example, the transmitting UE may transmit two consecutive SCIs (e.g., 2-stage SCI) on the PSSCH and/or PSSCH to the receiving UE. The receiving UE may decode the two consecutive SCIs (e.g., 2-stage SCI) to receive the PSSCH from the transmitting UE. For example, when SCI configuration fields are divided into two groups in consideration of a (relatively) large SCI payload size, SCI including a first SCI configuration field group is referred to as first SCI. SCI including a second SCI configuration field group may be referred to as second SCI. For example, the transmitting UE may transmit the first SCI to the receiving UE on the PSSCH. For example, the transmitting UE may transmit the second SCI to the receiving UE on the PSSCH and/or PSSCH. For example, the second SCI may be transmitted to the receiving UE on an (independent) PSSCH or on a PSSCH in which the second SCI is piggybacked to data. For example, the two consecutive SCIs may be applied to different transmissions (e.g., unicast, broadcast, or group-cast).

For example, the transmitting UE may transmit all or part of the following information to the receiving UE by SCI. For example, the transmitting UE may transmit all or part of the following information to the receiving UE by first SCI and/or second SCI.

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PSSCH-related and/or PSCCH-related resource allocation information, for example, the positions/number of time/frequency resources, resource reservation information (e.g. a periodicity), and/or

an SL channel state information (CSI) report request indicator or SL (L1) RSRP (and/or SL (L1) reference signal received quality (RSRQ) and/or SL (L1) received signal strength indicator (RSSI)) report request indicator, and/or

an SL CSI transmission indicator (on PSSCH) (or SL (L1) RSRP (and/or SL (L1) RSRQ and/or SL (L1) RSSI) information transmission indicator), and/or

MCS information, and/or

transmission power information, and/or

L1 destination ID information and/or L1 source ID information, and/or

SL HARQ process ID information, and/or

new data indicator (NDI) information, and/or

redundancy version (RV) information, and/or

QoS information (related to transmission traffic/packet), for example, priority information, and/or

an SL CSI-RS transmission indicator or information about the number of SL CSI-RS antenna ports (to be transmitted);

location information about a transmitting UE or location (or distance area) information about a target receiving UE (requested to transmit an SL HARQ feedback), and/or

RS (e.g., DMRS or the like) information related to decoding and/or channel estimation of data transmitted on a PSSCH, for example, information related to a pattern of (time-frequency) mapping resources of the DMRS, rank information, and antenna port index information.

For example, the first SCI may include information related to channel sensing. For example, the receiving UE may decode the second SCI using the PSSCH DMRS. A polar code used for the PDCCH may be applied to the second SCI. For example, the payload size of the first SCI may be equal for unicast, groupcast and broadcast in a resource pool. After decoding the first SCI, the receiving UE does not need to perform blind decoding on the second SCI. For example, the first SCI may include scheduling information about the second SCI.

In various embodiments of the present disclosure, since the transmitting UE may transmit at least one of the SCI, the first SCI, or the second SCI to the receiving UE on the PSCCH, the PSCCH may be replaced with at least one of the SCI, the first SCI, or the second SC. Additionally or alternatively, for example, the SCI may be replaced with at least one of the PSCCH, the first SCI, or the second SCI. Additionally or alternatively, for example, since the transmitting UE may transmit the second SCI to the receiving UE on the PSSCH, the PSSCH may be replaced with the second SCI.

FIG. 11 illustrates an ITS protocol stack. An intelligent transport systems (ITS) packet used for V2X communication is transmitted to a network layer (or network and protocol layer) through an access layer. The ITS packet may be transmitted via various paths as follows according to a combination of networking protocols. In this regard, the ITS protocol stack used for V2X communication is illustrated in FIG. 12. When the ITS packet is transmitted through a GeoNetworking protocol as in path number 1 or 2, an overall packet structure is defined as follows according to an access layer technology.

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Here, respective used methods are defined in a GeoNetworking common header and a GN-Data.request service primitive according to access layer technology based on ITS-G5 and LTE-V2X. Currently, when an ITS packet/V2X-related packet is transmitted according to the LTE-V2X access layer technology, a GeoNetworking protocol may not distinguish between communication over LTE-V2X PC5 or communication over an LTE-V2X Uu-interface. Therefore, a problem may arise when the GeoNetworking protocol transmits the V2X-related packet received from an access layer through a basic transport protocol (BTP) rather than a transmission control protocol (TCP)/Internet protocol (IP) although the V2X-related packet is a packet related to an LTE-V2X Uu interface. Hereinafter, embodiments for solving this problem are disclosed.

An operation method of a first UE according to an embodiment includes receiving a V2X-related packet from one of a second UE and a BS; and acquiring, by the first UE, a V2X-related message based on parsing of the V2X-related packet. The V2X-related packet includes header information indicating whether the V2X-related packet is an LTE-V2X PC5-related packet or an LTE-V2X Uu interface-related packet. Here, a network layer of the first UE may determine whether to process the V2X-related packet through a BTP or through a TCP, based on the header information. The V2X-related packet is transmitted to a higher layer through a GeoNetworking protocol, and the header information may be a Traffic Class field of a GeoNetworking common header.

In more detail, the traffic class field of the GeoNetworking common header has a traffic class (TC) ID value, and the TC ID is encoded as shown in Table 5 below.

TABLE 5

Field name	Type	Description
TC ID	6-bit unsigned integer	TC ID as specified in the media-dependent part of GeoNetworking corresponding to the interface over which the packet will be transmitted, e.g. for ITS-G5 and for LTE-V2X. Length: 6 bits

The TC field has a 6-bit unsigned integer value called TC ID and may use each value according to the access layer technology. The TC field of a GeoNetworking common header of LTE-V2X is used as shown in Table 6 below. LTE-V2X may express the mapping of packets of CAMs and DENMs by mapping PPPP levels and TC values.

TABLE 6

TC	PPPP	Intended Use
0	2	High priority DENMs
1	4	Normal DENMs
2	5	CAMs
3	7	Forwarded DENMs and other low priority messages

Here, the TC field of the GeoNetworking common header of LTE-V2X may be extended as shown in Table 7 below to express packets of LTE-V2X transmitted through a Uu-interface. In the table below, TC 4 to TC 9 have been added. However, all of the TCs should not necessarily be included and only some thereof may be added to the existing TCs as necessary.

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TABLE 7

TC	PPPP	Intended Use
0	2	High priority DENMs
1	4	Normal DENMs
2	5	CAMs
3	7	Forwarded DENMs and other low priority messages
4	2	High priority DENMs over Uu-interface
5	4	Normal DENMs over Uu-interface
6	5	CAMs over Uu-interface
7	7	Forwarded DENMs and other low priority messages over Uu-interface
8	1	VAMs over LTE-V2X PC5 (short range)
9	1	VAMs over LTE-V2X Uu-interface (long range)
10-63	N/A	Reserved for the future use

Hereinafter, methods of indicating one of an LTE-V2X PC5 interface and an LTE-V2X Uu interface will be described. The methods described below may be used independently but may have to be used together with the above-described TC field related method.

An embodiment proposes a method of extending itsGnIfType, which is an integer defining the interface type of a GeoNetworking protocol. That is, itsGnIfType, which is an integer defining the interface type of the GeoNetworking protocol, may indicate which one of the LTE-V2X PC5 interface and the LTE-V2X Uu interface the interface type is. This may be expressed as in Table 8 below.

TABLE 8

GeoNetworking protocol constant	Default/initial value	Comment
itsGnIfType	Unspecified (0) ITS-G5 (1) LTE-V2X PC5 (2) LTE-V2X Uu-interface (3)	Indicates type of interface

An ASN.1 module related to itsGnIfType may indicate which one of the LTE-V2X PC5 interface and the LTE-V2X Uu interface the interface type is. That is, according to the extension of the itsGnIfType constant, the definition in an ASN.1 module may be extended as shown in Table 9 below (ASN.1 encoding of a GeoNetworking MIB).

TABLE 9

itsGnIfType OBJECT-TYPE
SYNTAX INTEGER {
unspecified(0),
its-g5(1),
lte-v2x-pc5(2)
lte-v2x-uu-interface(3)
}
MAX-ACCESS read-only
STATUS current
DESCRIPTION
"ITS interface type."
::= { itsGnConfig 5 }

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As another example, the V2X-related packet may be generated based on a communication profile of GN-DATA.request indicating which one of the LTE-V2X PC5 interface and the LTE-V2X Uu interface the interface type is. Here, the V2X-related packet may be generated based on a GN-DATA.request service primitive including a range parameter and a DirectCommunication parameter. In addition, the range parameter may indicate one of a short range and a long range, and the DirectCommunication parameter may indicate one of direct communication and indirect communication.

Considering this from the perspective of a transmitting side, an ITS transport protocol entity may deliver a GN-DATA.request service primitive to a GeoNetworking protocol entity in order to transmit a packet including LTE-V2X communication-related interface information (Uu or PC5) to another device. The GeoNetworking protocol entity may generate a GeoNetworking packet related to the LTE-V2X communication-related interface information (Uu or PC5) based on the delivered GN-DATA.request service primitive. The generated GeoNetworking packet may be transmitted to another device, and the other device may identify through which LTE-V2X communication-related interface a signal or packet is received based on the GeoNetworking packet.

That is, the GN-DATA.request service primitive is used by the ITS transport protocol entity to request sending of the GeoNetworking packet. In other words, the ITS transport protocol entity may transmit a request related to delivery of the GeoNetworking packet to the GeoNetworking protocol entity through the GN-DATA.request service primitive. Upon receiving the GN-DATA.request service primitive from the ITS transport protocol entity, the GeoNetworking protocol delivers the GeoNetworking packet to a link layer (LL) protocol entity via an ITS access-network service access point (IN_SAP).

An embodiment proposes a method of extending a parameter of the GN-DATA.request service primitive or adding a new parameter in order to support the LTE-V2X Uu interface as the access layer technology.

The method of extending a communication profile parameter is as follows.

The communication profile parameter of GN-DATA.request determines the LL protocol entity.

A communication profile extension plan is as follows.

- 1) Original: Unspecified, ITS-G5, LTE-V2X
- 2) Extension: Unspecified, ITS-G5, LTE-V2X PC5 (short range), LTE-V2X Uu interface (long range)

The method of adding the new parameter is as follows.

A method of supporting LTE-V2X Uu interface access technology by adding the new parameter to the GN-DATA.request service primitive is proposed. By newly adding a range parameter and a DirectCommunication parameter, whether an interface type is a Uu interface or a PC5 interface when the communication profile parameter is LTE-V2X is indicated.

TABLE 10

```

GN-DATA.request (
    Upper protocol entity,
    Packet transport type,
    Destination address,
    Communication profile,
    Security profile, (optional)
    ITS-AID length, (optional)
    ITS-AID, (optional)

```


TABLE 10-continued

```

Security permissions length, (optional)
Security permissions, (optional)
Security context information, (optional)
Security target ID list length, (optional)
Security target ID list, (optional)
Maximum packet lifetime, (optional)
Repetition interval, (optional)
Maximum repetition time, (optional)
Maximum hop limit, (optional)
Traffic class,
Range (optional),
DirectCommunication (optional)
Length,
Data
)

```

FIG. 13 is a flowchart related to an embodiment. For a description of each part of the flowchart, reference is made to the above description.

FIG. 14 illustrates a method of acquiring interface or protocol information related to LTE-V2X communication associated with the above-described embodiments. A first layer may receive a packet. The first layer may parse a plurality of fields included in the received packet (S901). The first layer may deliver a predetermined field or header (or GeoNetworking Basic Header or GeoNetworking common header) related to a second layer to the second layer. The second layer may acquire first information based on the predetermined field or header received from the first layer (S903). The first information may be information as to whether the packet is a packet related to communication over an LTE-V2X PC5 interface or a packet related to communication over an LTE-V2X Uu interface.

Alternatively, the packet may include the plurality of fields described above, and the interface or protocol information related to LTE-V2X communication may be included in a specific header field. Alternatively, the first layer may be an access layer, and the second layer may be a layer related to a GeoNetworking protocol. In this case, the specific header field may be a GeoNetworking common header, and the second layer may acquire the interface or protocol information related to LTE-V2X communication from the Traffic Class field of the GeoNetworking common header.

Through configuration in each layer as described above, a problem that the GeoNetworking protocol incorrectly transmits a TCP/IP packet used for the LTE-V2X Uu interface through the BTP may be solved.

A first UE related to the above-described embodiment may include at least one processor; and at least one computer memory operably connectable to the at least one processor and storing instructions that, when executed, cause the at least one processor to perform operations. The operations may include receiving a V2X-related packet from one of a second UE and a BS; and acquiring a V2X-related message based on parsing of the V2X-related packet. The V2X-related packet may include header information indicating whether the V2X-related packet is an LTE-V2X PC5-related packet or an LTE-V2X Uu interface-related packet.

In a processor for performing operations for a first UE, the operations may include receiving a V2X-related packet from one of a second UE and a BS; and acquiring a V2X-related message based on parsing of the V2X-related packet. The V2X-related packet may include header information indicating whether the V2X-related packet is an LTE-V2X PC5-related packet or an LTE-V2X Uu interface-related packet.

In a non-volatile computer readable storage medium storing at least one computer program including instructions that, when executed by at least one processor, cause the at least one processor to perform operations for a UE, the operations may include receiving a V2X-related packet from one of a second UE and a BS; and acquiring a V2X-related message based on parsing of the V2X-related packet. The V2X-related packet may include header information indicating whether the V2X-related packet is an LTE-V2X PC5-related packet or an LTE-V2X Uu interface-related packet.

Examples of Communication Systems Applicable to the Present Disclosure

The various descriptions, functions, procedures, proposals, methods, and/or operational flowcharts of the present disclosure described in this document may be applied to, without being limited to, a variety of fields requiring wireless communication/connection (e.g., 5G) between devices.

Hereinafter, a description will be given in more detail with reference to the drawings. In the following drawings/description, the same reference symbols may denote the same or corresponding hardware blocks, software blocks, or functional blocks unless described otherwise.

FIG. 15 illustrates a communication system 1 applied to the present disclosure.

Referring to FIG. 15, a communication system 1 applied to the present disclosure includes wireless devices, BSs, and a network. Herein, the wireless devices represent devices performing communication using RAT (e.g., 5G NR or LTE) and may be referred to as communication/radio/5G devices. The wireless devices may include, without being limited to, a robot 100a, vehicles 100b-1 and 100b-2, an extended reality (XR) device 100c, a hand-held device 100d, a home appliance 100e, an Internet of things (IoT) device 100f, and an artificial intelligence (AI) device/server 400. For example, the vehicles may include a vehicle having a wireless communication function, an autonomous driving vehicle, and a vehicle capable of performing communication between vehicles. Herein, the vehicles may include an unmanned aerial vehicle (UAV) (e.g., a drone). The XR device may include an augmented reality (AR)/virtual reality (VR)/mixed reality (MR) device and may be implemented in the form of a head-mounted device (HMD), a head-up display (HUD) mounted in a vehicle, a television, a smartphone, a computer, a wearable device, a home appliance device, a digital signage, a vehicle, a robot, etc. The hand-held device may include a smartphone, a smartpad, a wearable device (e.g., a smartwatch or a smartglasses), and a computer (e.g., a notebook). The home appliance may include a TV, a refrigerator, and a washing

machine. The IoT device may include a sensor and a smartmeter. For example, the BSs and the network may be implemented as wireless devices and a specific wireless device **200a** may operate as a BS/network node with respect to other wireless devices.

The wireless devices **100a** to **100f** may be connected to the network **300** via the BSs **200**. An AI technology may be applied to the wireless devices **100a** to **100f** and the wireless devices **100a** to **100f** may be connected to the AI server **400** via the network **300**. The network **300** may be configured using a 3G network, a 4G (e.g., LTE) network, or a 5G (e.g., NR) network. Although the wireless devices **100a** to **100f** may communicate with each other through the BSs **200**/network **300**, the wireless devices **100a** to **100f** may perform direct communication (e.g., sidelink communication) with each other without passing through the BSs/network. For example, the vehicles **100b-1** and **100b-2** may perform direct communication (e.g., V2V/V2X communication). The IoT device (e.g., a sensor) may perform direct communication with other IoT devices (e.g., sensors) or other wireless devices **100a** to **100f**.

Wireless communication/connections **150a**, **150b**, or **150c** may be established between the wireless devices **100a** to **100f**/BS **200**, or BS **200**/BS **200**. Herein, the wireless communication/connections may be established through various RATs (e.g., 5G NR) such as UL/DL communication **150a**, sidelink communication **150b** (or, D2D communication), or inter BS communication (e.g., relay, integrated access backhaul (IAB)). The wireless devices and the BSs/the wireless devices may transmit/receive radio signals to/from each other through the wireless communication/connections **150a** and **150b**. For example, the wireless communication/connections **150a** and **150b** may transmit/receive signals through various physical channels. To this end, at least a part of various configuration information configuring processes, various signal processing processes (e.g., channel encoding/decoding, modulation/demodulation, and resource mapping/demapping), and resource allocating processes, for transmitting/receiving radio signals, may be performed based on the various proposals of the present disclosure.

Examples of Wireless Devices Applicable to the Present Disclosure

FIG. 16 illustrates wireless devices applicable to the present disclosure.

Referring to FIG. 16, a first wireless device **100** and a second wireless device **200** may transmit radio signals through a variety of RATs (e.g., LTE and NR). Herein, {the first wireless device **100** and the second wireless device **200**} may correspond to {the wireless device **100x** and the BS **200**} and/or {the wireless device **100x** and the wireless device **100x**} of FIG. 15.

The first wireless device **100** may include one or more processors **102** and one or more memories **104** and additionally further include one or more transceivers **106** and/or one or more antennas **108**. The processor(s) **102** may control the memory(s) **104** and/or the transceiver(s) **106** and may be configured to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. For example, the processor(s) **102** may process information within the memory(s) **104** to generate first information/signals and then transmit radio signals including the first information/signals through the transceiver(s) **106**. The processor(s) **102** may receive radio signals including second information/signals through the transceiver **106** and then store information obtained by processing the second information/signals in the memory(s)

104. The memory(s) **104** may be connected to the processor(s) **102** and may store a variety of information related to operations of the processor(s) **102**. For example, the memory(s) **104** may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) **102** or for performing the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. Herein, the processor(s) **102** and the memory(s) **104** may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) **106** may be connected to the processor(s) **102** and transmit and/or receive radio signals through one or more antennas **108**. Each of the transceiver(s) **106** may include a transmitter and/or a receiver. The transceiver(s) **106** may be interchangeably used with Radio Frequency (RF) unit(s). In the present disclosure, the wireless device may represent a communication modem/circuit/chip.

The second wireless device **200** may include one or more processors **202** and one or more memories **204** and additionally further include one or more transceivers **206** and/or one or more antennas **208**. The processor(s) **202** may control the memory(s) **204** and/or the transceiver(s) **206** and may be configured to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. For example, the processor(s) **202** may process information within the memory(s) **204** to generate third information/signals and then transmit radio signals including the third information/signals through the transceiver(s) **206**. The processor(s) **202** may receive radio signals including fourth information/signals through the transceiver(s) **106** and then store information obtained by processing the fourth information/signals in the memory(s) **204**. The memory(s) **204** may be connected to the processor(s) **202** and may store a variety of information related to operations of the processor(s) **202**. For example, the memory(s) **204** may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) **202** or for performing the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. Herein, the processor(s) **202** and the memory(s) **204** may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) **206** may be connected to the processor(s) **202** and transmit and/or receive radio signals through one or more antennas **208**. Each of the transceiver(s) **206** may include a transmitter and/or a receiver. The transceiver(s) **206** may be interchangeably used with RF unit(s). In the present disclosure, the wireless device may represent a communication modem/circuit/chip.

Hereinafter, hardware elements of the wireless devices **100** and **200** will be described more specifically. One or more protocol layers may be implemented by, without being limited to, one or more processors **102** and **202**. For example, the one or more processors **102** and **202** may implement one or more layers (e.g., functional layers such as PHY, MAC, RLC, PDCP, RRC, and SDAP). The one or more processors **102** and **202** may generate one or more Protocol Data Units (PDUs) and/or one or more service data unit (SDUs) according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. The one or more processors **102** and **202** may generate messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. The one or more proces-

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sors **102** and **202** may generate signals (e.g., baseband signals) including PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document and provide the generated signals to the one or more transceivers **106** and **206**. The one or more processors **102** and **202** may receive the signals (e.g., baseband signals) from the one or more transceivers **106** and **206** and acquire the PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document.

The one or more processors **102** and **202** may be referred to as controllers, microcontrollers, microprocessors, or microcomputers. The one or more processors **102** and **202** may be implemented by hardware, firmware, software, or a combination thereof. As an example, one or more application specific integrated circuits (ASICs), one or more digital signal processors (DSPs), one or more digital signal processing devices (DSPDs), one or more programmable logic devices (PLDs), or one or more field programmable gate arrays (FPGAs) may be included in the one or more processors **102** and **202**. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be implemented using firmware or software and the firmware or software may be configured to include the modules, procedures, or functions. Firmware or software configured to perform the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be included in the one or more processors **102** and **202** or stored in the one or more memories **104** and **204** so as to be driven by the one or more processors **102** and **202**. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be implemented using firmware or software in the form of code, commands, and/or a set of commands.

The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** and store various types of data, signals, messages, information, programs, code, instructions, and/or commands. The one or more memories **104** and **204** may be configured by read-only memories (ROMs), random access memories (RAMs), electrically erasable programmable read-only memories (EPROMs), flash memories, hard drives, registers, cash memories, computer-readable storage media, and/or combinations thereof. The one or more memories **104** and **204** may be located at the interior and/or exterior of the one or more processors **102** and **202**. The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** through various technologies such as wired or wireless connection.

The one or more transceivers **106** and **206** may transmit user data, control information, and/or radio signals/channels, mentioned in the methods and/or operational flowcharts of this document, to one or more other devices. The one or more transceivers **106** and **206** may receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document, from one or more other devices. For example, the one or more transceivers **106** and **206** may be connected to the one or more processors **102** and **202** and transmit and receive radio signals. For example, the one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may transmit user data,

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control information, or radio signals to one or more other devices. The one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may receive user data, control information, or radio signals from one or more other devices. The one or more transceivers **106** and **206** may be connected to the one or more antennas **108** and **208** and the one or more transceivers **106** and **206** may be configured to transmit and receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document, through the one or more antennas **108** and **208**. In this document, the one or more antennas may be a plurality of physical antennas or a plurality of logical antennas (e.g., antenna ports). The one or more transceivers **106** and **206** may convert received radio signals/channels etc. from RF band signals into baseband signals in order to process received user data, control information, radio signals/channels, etc. using the one or more processors **102** and **202**. The one or more transceivers **106** and **206** may convert the user data, control information, radio signals/channels, etc. processed using the one or more processors **102** and **202** from the base band signals into the RF band signals. To this end, the one or more transceivers **106** and **206** may include (analog) oscillators and/or filters.

Examples of a Vehicle or an Autonomous Driving Vehicle Applicable to the Present Disclosure

FIG. 17 illustrates a vehicle or an autonomous driving vehicle applied to the present disclosure. The vehicle or autonomous driving vehicle may be implemented by a mobile robot, a car, a train, a manned/unmanned aerial vehicle (AV), a ship, etc.

Referring to FIG. 17, a vehicle or autonomous driving vehicle **100** may include an antenna unit **108**, a communication unit **110**, a control unit **120**, a driving unit **140a**, a power supply unit **140b**, a sensor unit **140c**, and an autonomous driving unit **140d**. The antenna unit **108** may be configured as a part of the communication unit **110**.

The communication unit **110** may transmit and receive signals (e.g., data and control signals) to and from external devices such as other vehicles, BSs (e.g., gNBs and road side units), and servers. The control unit **120** may perform various operations by controlling elements of the vehicle or the autonomous driving vehicle **100**. The control unit **120** may include an ECU. The driving unit **140a** may cause the vehicle or the autonomous driving vehicle **100** to drive on a road. The driving unit **140a** may include an engine, a motor, a powertrain, a wheel, a brake, a steering device, etc. The power supply unit **140b** may supply power to the vehicle or the autonomous driving vehicle **100** and include a wired/wireless charging circuit, a battery, etc. The sensor unit **140c** may acquire a vehicle state, ambient environment information, user information, etc. The sensor unit **140c** may include an inertial measurement unit (IMU) sensor, a collision sensor, a wheel sensor, a speed sensor, a slope sensor, a weight sensor, a heading sensor, a position module, a vehicle forward/backward sensor, a battery sensor, a fuel sensor, a tire sensor, a steering sensor, a temperature sensor, a humidity sensor, an ultrasonic sensor, an illumination sensor, a pedal position sensor, etc. The autonomous driving unit **140d** may implement technology for maintaining a lane on which a vehicle is driving, technology for automatically adjusting speed, such as adaptive cruise control, technology for autonomously driving along a determined path, technology for driving by automatically setting a path if a destination is set, and the like.

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For example, the communication unit **110** may receive map data, traffic information data, etc. from an external server. The autonomous driving unit **140d** may generate an autonomous driving path and a driving plan from the obtained data. The control unit **120** may control the driving unit **140a** such that the vehicle or the autonomous driving vehicle **100** may move along the autonomous driving path according to the driving plan (e.g., speed/direction control). In the middle of autonomous driving, the communication unit **110** may aperiodically/periodically acquire recent traffic information data from the external server and acquire surrounding traffic information data from neighboring vehicles. In the middle of autonomous driving, the sensor unit **140c** may obtain a vehicle state and/or surrounding environment information. The autonomous driving unit **140d** may update the autonomous driving path and the driving plan based on the newly obtained data/information. The communication unit **110** may transfer information about a vehicle position, the autonomous driving path, and/or the driving plan to the external server. The external server may predict traffic information data using AI technology, etc., based on the information collected from vehicles or autonomous driving vehicles and provide the predicted traffic information data to the vehicles or the autonomous driving vehicles.

Examples of a Vehicle and AR/VR Applicable to the Present Disclosure

FIG. **18** illustrates a vehicle applied to the present disclosure. The vehicle may be implemented as a transport means, an aerial vehicle, a ship, etc.

Referring to FIG. **18**, a vehicle **100** may include a communication unit **110**, a control unit **120**, a memory unit **130**, an I/O unit **140a**, and a positioning unit **140b**.

The communication unit **110** may transmit and receive signals (e.g., data and control signals) to and from external devices such as other vehicles or BSs. The control unit **120** may perform various operations by controlling constituent elements of the vehicle **100**. The memory unit **130** may store data/parameters/programs/code/commands for supporting various functions of the vehicle **100**. The I/O unit **140a** may output an AR/VR object based on information within the memory unit **130**. The I/O unit **140a** may include an HUD. The positioning unit **140b** may acquire information about the position of the vehicle **100**. The position information may include information about an absolute position of the vehicle **100**, information about the position of the vehicle **100** within a traveling lane, acceleration information, and information about the position of the vehicle **100** from a neighboring vehicle. The positioning unit **140b** may include a GPS and various sensors.

As an example, the communication unit **110** of the vehicle **100** may receive map information and traffic information from an external server and store the received information in the memory unit **130**. The positioning unit **140b** may obtain the vehicle position information through the GPS and various sensors and store the obtained information in the memory unit **130**. The control unit **120** may generate a virtual object based on the map information, traffic information, and vehicle position information and the I/O unit **140a** may display the generated virtual object in a window in the vehicle (**1410** and **1420**). The control unit **120** may determine whether the vehicle **100** normally drives within a traveling lane, based on the vehicle position information. If the vehicle **100** abnormally exits from the traveling lane, the control unit **120** may display a warning on the window in the vehicle through the I/O unit **140a**. In addition, the control unit **120** may broadcast a warning message regarding driving abnormality to neighboring vehicles through the commu-

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nication unit **110**. According to situation, the control unit **120** may transmit the vehicle position information and the information about driving/vehicle abnormality to related organizations.

Examples of an XR Device Applicable to the Present Disclosure

FIG. **19** illustrates an XR device applied to the present disclosure. The XR device may be implemented by an HMD, an HUD mounted in a vehicle, a television, a smartphone, a computer, a wearable device, a home appliance, a digital signage, a vehicle, a robot, etc.

Referring to FIG. **19**, an XR device **100a** may include a communication unit **110**, a control unit **120**, a memory unit **130**, an I/O unit **140a**, a sensor unit **140b**, and a power supply unit **140c**.

The communication unit **110** may transmit and receive signals (e.g., media data and control signals) to and from external devices such as other wireless devices, hand-held devices, or media servers. The media data may include video, images, and sound. The control unit **120** may perform various operations by controlling constituent elements of the XR device **100a**. For example, the control unit **120** may be configured to control and/or perform procedures such as video/image acquisition, (video/image) encoding, and metadata generation and processing. The memory unit **130** may store data/parameters/programs/code/commands needed to drive the XR device **100a**/generate XR object. The I/O unit **140a** may obtain control information and data from the exterior and output the generated XR object. The I/O unit **140a** may include a camera, a microphone, a user input unit, a display unit, a speaker, and/or a haptic module. The sensor unit **140b** may obtain an XR device state, surrounding environment information, user information, etc. The sensor unit **140b** may include a proximity sensor, an illumination sensor, an acceleration sensor, a magnetic sensor, a gyro sensor, an inertial sensor, an RGB sensor, an IR sensor, a fingerprint recognition sensor, an ultrasonic sensor, a light sensor, a microphone and/or a radar. The power supply unit **140c** may supply power to the XR device **100a** and include a wired/wireless charging circuit, a battery, etc.

For example, the memory unit **130** of the XR device **100a** may include information (e.g., data) needed to generate the XR object (e.g., an AR/VR/MR object). The I/O unit **140a** may receive a command for manipulating the XR device **100a** from a user and the control unit **120** may drive the XR device **100a** according to a driving command of a user. For example, when a user desires to watch a film or news through the XR device **100a**, the control unit **120** transmits content request information to another device (e.g., a hand-held device **100b**) or a media server through the communication unit **130**. The communication unit **130** may download/stream content such as films or news from another device (e.g., the hand-held device **100b**) or the media server to the memory unit **130**. The control unit **120** may control and/or perform procedures such as video/image acquisition, (video/image) encoding, and metadata generation/processing with respect to the content and generate/output the XR object based on information about a surrounding space or a real object obtained through the I/O unit **140a**/sensor unit **140b**.

The XR device **100a** may be wirelessly connected to the hand-held device **100b** through the communication unit **110** and the operation of the XR device **100a** may be controlled by the hand-held device **100b**. For example, the hand-held device **100b** may operate as a controller of the XR device **100a**. To this end, the XR device **100a** may obtain infor-

mation about a 3D position of the hand-held device **100b** and generate and output an XR object corresponding to the hand-held device **100b**.

Examples of a Robot Applicable to the Present Disclosure

FIG. 20 illustrates a robot applied to the present disclosure. The robot may be categorized into an industrial robot, a medical robot, a household robot, a military robot, etc., according to a used purpose or field.

Referring to FIG. 20, a robot **100** may include a communication unit **110**, a control unit **120**, a memory unit **130**, an I/O unit **140a**, a sensor unit **140b**, and a driving unit **140c**. Herein, the blocks **110** to **130/140a** to **140c** correspond to the blocks **110** to **130/140** of FIG. 16, respectively.

The communication unit **110** may transmit and receive signals (e.g., driving information and control signals) to and from external devices such as other wireless devices, other robots, or control servers. The control unit **120** may perform various operations by controlling constituent elements of the robot **100**. The memory unit **130** may store data/parameters/programs/code/commands for supporting various functions of the robot **100**. The I/O unit **140a** may obtain information from the exterior of the robot **100** and output information to the exterior of the robot **100**. The I/O unit **140a** may include a camera, a microphone, a user input unit, a display unit, a speaker, and/or a haptic module. The sensor unit **140b** may obtain internal information of the robot **100**, surrounding environment information, user information, etc. The sensor unit **140b** may include a proximity sensor, an illumination sensor, an acceleration sensor, a magnetic sensor, a gyro sensor, an inertial sensor, an IR sensor, a fingerprint recognition sensor, an ultrasonic sensor, a light sensor, a microphone, a radar, etc. The driving unit **140c** may perform various physical operations such as movement of robot joints. In addition, the driving unit **140c** may cause the robot **100** to travel on the road or to fly. The driving unit **140c** may include an actuator, a motor, a wheel, a brake, a propeller, etc.

Example of AI device to which the present disclosure is applied.

FIG. 21 illustrates an AI device applied to the present disclosure. The AI device may be implemented by a fixed device or a mobile device, such as a TV, a projector, a smartphone, a PC, a notebook, a digital broadcast terminal, a tablet PC, a wearable device, a Set Top Box (STB), a radio, a washing machine, a refrigerator, a digital signage, a robot, a vehicle, etc.

Referring to FIG. 21, an AI device **100** may include a communication unit **110**, a control unit **120**, a memory unit **130**, an I/O unit **140a/140b**, a learning processor unit **140c**, and a sensor unit **140d**.

The communication unit **110** may transmit and receive wired/radio signals (e.g., sensor information, user input, learning models, or control signals) to and from external devices such as other AI devices (e.g., **100x**, **200**, or **400** of FIG. 15) or an AI server (e.g., **400** of FIG. 15) using wired/wireless communication technology. To this end, the communication unit **110** may transmit information within the memory unit **130** to an external device and transmit a signal received from the external device to the memory unit **130**.

The control unit **120** may determine at least one feasible operation of the AI device **100**, based on information which is determined or generated using a data analysis algorithm or a machine learning algorithm. The control unit **120** may perform an operation determined by controlling constituent elements of the AI device **100**. For example, the control unit **120** may request, search, receive, or use data of the learning

processor unit **140c** or the memory unit **130** and control the constituent elements of the AI device **100** to perform a predicted operation or an operation determined to be preferred among at least one feasible operation. The control unit **120** may collect history information including the operation contents of the AI device **100** and operation feedback by a user and store the collected information in the memory unit **130** or the learning processor unit **140c** or transmit the collected information to an external device such as an AI server (**400** of FIG. 15). The collected history information may be used to update a learning model.

The memory unit **130** may store data for supporting various functions of the AI device **100**. For example, the memory unit **130** may store data obtained from the input unit **140a**, data obtained from the communication unit **110**, output data of the learning processor unit **140c**, and data obtained from the sensor unit **140**. The memory unit **130** may store control information and/or software code needed to operate/drive the control unit **120**.

The input unit **140a** may acquire various types of data from the exterior of the AI device **100**. For example, the input unit **140a** may acquire learning data for model learning, and input data to which the learning model is to be applied. The input unit **140a** may include a camera, a microphone, and/or a user input unit. The output unit **140b** may generate output related to a visual, auditory, or tactile sense. The output unit **140b** may include a display unit, a speaker, and/or a haptic module. The sensing unit **140** may obtain at least one of internal information of the AI device **100**, surrounding environment information of the AI device **100**, and user information, using various sensors. The sensor unit **140** may include a proximity sensor, an illumination sensor, an acceleration sensor, a magnetic sensor, a gyro sensor, an inertial sensor, an RGB sensor, an IR sensor, a fingerprint recognition sensor, an ultrasonic sensor, a light sensor, a microphone, and/or a radar.

The learning processor unit **140c** may learn a model consisting of artificial neural networks, using learning data. The learning processor unit **140c** may perform AI processing together with the learning processor unit of the AI server (**400** of FIG. 15). The learning processor unit **140c** may process information received from an external device through the communication unit **110** and/or information stored in the memory unit **130**. In addition, an output value of the learning processor unit **140c** may be transmitted to the external device through the communication unit **110** and may be stored in the memory unit **130**.

Industrial Availability

The above-described embodiments of the present disclosure are applicable to various mobile communication systems.

The invention claimed is:

1. A method comprising:

- establishing, by a first user equipment (UE) with a second UE, a connection related to UE to UE direct communication;
- receiving, by the first UE, a scheduling information from a base station;
- transmitting, by the first UE, at least one UE to UE direct communication-related first packet through a resource related to the scheduling information;
- receiving, by the first UE, a UE to UE direct communication-related second packet from one of the second UE and the base station; and
- acquiring, by the first UE, a UE to UE direct communication-related message based on parsing of the UE to UE direct communication-related second packet,

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wherein the UE to UE direct communication-related second packet includes header information indicating whether the UE to UE direct communication-related second packet is a long-term evolution (LTE)-UE to UE direct communication PC5-related packet or an LTE-UE to UE direct communication Uu interface-related packet,

wherein a network layer of the first UE determines whether to process the UE to UE direct communication-related second packet through a basic transport protocol (BTP) or through a transmission control protocol (TCP), based on the header information,

wherein the UE to UE direct communication-related second packet is transmitted to a higher layer through a GeoNetworking protocol,

wherein the header information is a traffic class field of a GeoNetworking common header.

2. The method of claim 1, wherein an integer, itsGnIfType, defining an interface type of the GeoNetworking protocol indicates which one of an LTE-UE to UE direct communication PC5 interface or an LTE-UE to UE direct communication Uu-interface the interface type is.

3. The method of claim 2, wherein an ASN.1 module related to the integer itsGnIfType indicates which one of the LTE-UE to UE direct communication PC5 interface and the LTE-UE to UE direct communication Uu-interface the interface type is.

4. The method of claim 3, wherein the UE to UE direct communication-related second packet is generated based on a communication profile of GN-DATA.request indicating which one of the LTE-UE to UE direct communication PC5 interface and the LTE-UE to UE direct communication Uu interface the interface type is.

5. The method of claim 3, wherein the UE to UE direct communication-related second packet is generated based on a GN-DATA.request service primitive including a range parameter and a DirectCommunication parameter.

6. The method of claim 5, wherein the range parameter indicates one of a short range and a long range, and the DirectCommunication parameter indicates one of direct communication and indirect communication.

7. The method of claim 1, wherein the UE to UE direct communication-related second packet is an intelligent transport systems (ITS) packet.

8. A first user equipment (UE) in a wireless communication system, the first UE comprising:

at least one processor; and

at least one computer memory operably connectable to the at least one processor and storing instructions that, when executed, cause the at least one processor to perform operations,

wherein the operations comprise:

receiving, by the first UE, a UE to UE direct communication-related second packet from one of a second UE and a base station; and

acquiring, by the first UE, a UE to UE direct communication-related message based on parsing of the UE to UE direct communication-related second packet, and

wherein the UE to UE direct communication-related second packet includes header information indicating whether the UE to UE direct communication-related second packet is a long-term evolution (LTE)-UE to UE direct communication PC5-related packet or an LTE-UE to UE direct communication Uu interface-related packet,

wherein a network layer of the first UE determines whether to process the UE to UE direct communication-related second packet through a basic transport

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protocol (BTP) or through a transmission control protocol (TCP), based on the header information,

wherein the UE to UE direct communication-related second packet is transmitted to a higher layer through a GeoNetworking protocol,

wherein the header information is a traffic class field of a GeoNetworking common header.

9. A processor for performing operations for a first user equipment (UE) in a wireless communication system, the operations comprising:

establishing, by the first UE with a second UE, a connection related to UE to UE direct communication;

receiving, by the first UE, a scheduling information from a base station;

transmitting, by the first UE, at least one UE to UE direct communication-related first packet through a resource related to the scheduling information;

receiving, by the first UE, a UE to UE direct communication-related second packet from one of the second UE and the base station; and

acquiring, by the first UE, a UE to UE direct communication-related message based on parsing of the UE to UE direct communication-related second packet,

wherein the UE to UE direct communication-related second packet includes header information indicating whether the UE to UE direct communication-related second packet is a long-term evolution (LTE)-UE to UE direct communication PC5-related packet or an LTE-UE to UE direct communication Uu interface-related packet,

wherein a network layer of the first UE determines whether to process the UE to UE direct communication-related second packet through a basic transport protocol (BTP) or through a transmission control protocol (TCP), based on the header information,

wherein the UE to UE direct communication-related second packet is transmitted to a higher layer through a GeoNetworking protocol,

wherein the header information is a traffic class field of a GeoNetworking common header.

10. A non-transitory computer readable storage medium storing at least one computer program including instructions that, when executed by at least one processor, cause the at least one processor to perform operations for a first user equipment (UE), the operations comprising:

establishing, by the first UE with a second UE, a connection related to UE to UE direct communication;

receiving, by the first UE, a scheduling information from a base station;

transmitting, by the first UE, at least one UE to UE direct communication-related first packet through a resource related to the scheduling information;

receiving, by the first UE, a UE to UE direct communication-related second packet from one of the second UE and a base station; and

acquiring, by the first UE, a UE to UE direct communication-related message based on parsing of the UE to UE direct communication-related second packet,

wherein the UE to UE direct communication-related second packet includes header information indicating whether the UE to UE direct communication-related second packet is a long-term evolution (LTE)-UE to UE direct communication PC5-related packet or an LTE-UE to UE direct communication Uu interface-related packet,

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wherein a network layer of the first UE determines whether to process the UE to UE direct communication-related second packet through a basic transport protocol (BTP) or through a transmission control protocol (TCP), based on the header information, 5
wherein the UE to UE direct communication-related second packet is transmitted to a higher layer through a GeoNetworking protocol,
wherein the header information is a traffic class field of a GeoNetworking common header. 10

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